Scope of Livestock in Indian Economy Livestock Census, Trends in Livestock Production

Livestock production performance has been more impressive than that of food grain production. Milk, egg, meat, and fish showed impressive growth rates of 5to10%.The minimum targeted growth rate for attaining self sufficiency in milk, fish, meat and egg by 2001 AD are 5.54, 6.25, and 5.54 % per annum respectively.
Livestock represents the only way in which the natural vegetation that covers large parts of India can be converted in to products that can be used by man. It provides drought power and manure to the crop enterprise and this in turn provides feed and fodder. The value of out put from the livestock sector was Rupees 79684 crores in 1994-95 which was 9.3% the Total (GDP).

Fortunately India is blessed with a tremendous livestock wealth. It has the largest population of cattle and buffalo in the world and its breeds are admired for heat tolerance and inherent resistance to diseases and ability to thrive under different climatic condition.
The cattle population of India is very large. According 1991 census the cattle population was estimated at 467.9 million this comprised of 203.1 million catties, 83.1 million buffaloes 50.7million sheep, 115.3 million goats and 12.1 million pigs. The others were estimated at 3.6 million. The poultry population constituted a 400 million.

Milk production:

India ranks first with the average milk production of 78 million tons per annum. This has been the achievement of 70 million dairy farmers and also through the striated efforts of the animal husbandry practices, cattle cross breeding projects and cooperative dairy farming. It is worthwhile to mention that the per capita availability of milk to the lacto vegetarian Indians is estimated at 214 grams per day. It has been the only source of sufficient energy, minerals, vitamins and animal proteins. A 60% of the total milk production enters in. to the market in the form of dahi, butter, ghee, khoa and shrikhand. Besides this the conventional dairy products including milk powder, Ice cream and cheese are also manufactured. During last 20 years the supply f milk has been possible in sufficient quantities through the pasteurization plants and chilling units.

Animal draught power*:*

The bullock pair may be regarded as the backbone of Indian Agriculture. Though the animal draught power does not relate with human nutrition directly. Indirectly it contributes in the production of food grains; the renowned draught animals (cattle and buffaloes) include Khillar, Amrit mahal, Hallikar, Red kandhari, Ongole, Malvi, Rathi, Nagore, Neman, Hariyana, Gir, and Deoni. There are about 86 million draft animals, which comprise of 76 million bullocks. 8 million buffaloes, 1 million camels and donkeys. The horsepower obtained from 1 bullock is equivalent 0.75 H.P.
Mechanization in Agriculture has been to the tune of 20% only. Whereas 80% of the agriculture/farm operations are done by bullock drawn .implements. It is estimated that 40,000 mega watts, of Energy (Traction power) is made available through the use of draft animals and the value of this has been estimated Rs. 5000/- crores.

Meat production:

Flesh foods are rich in protein and are good sources of vitamin B12 which is absent in plant food. India's meat production is hardly 2% (4.08 million tons) of the global meat production 209.31 million tons in 1995. Out of total meat produce in India 54% is from mutton and chevon, 26% from beef 13% from chicken and 7% from pork. Even though 70% of India’s populations consume meat the per capita availability of meat is less than 5 kg per year. As compared to worlds average of 14 kg per year Broiler production in India is recent one rearing poultry for meat purposes started only in seventies, but the growth is significant Broiler production which was only 4 million in 1971 increased to around 215 million in 1991.
The poultry industry has achieved a spectacular growth during last thirty years. The 24'billion eggs produced in 1991 represented 13 fold increase compared to 1951.
With the annual production of 27 billion eggs (1995-96) India stands fifth in world. The government has promoted the poultry development through intensive poultry development project (IPDP) launched in third five years plan, (1969-74). Improved breeds like RIR, WLH and Australia. The per capita

availability of eggs in India is only 30 per annum as against the ICMR recommendation of 180 per year.

Fish production:

Fish is a cheap source of animal protein and a good source of calcium. The fish production of India has risen to 4.95 million tons in 1995-96. The per capita availability of fish in 1996 was 5.4 kg whereas the ICMR recommendation for total meat including fish is 10.95 kg per annum.

Farm yard manure for organic farming:

A minimum of 10-20 kg dung is obtained on an average from every cow or buffalo. This is an excellent source of F.Y.M. or compost manure. This is badly needed to improve the inherent soil fertility, and to have the extended manorial effect on the crops parts. Dung cakes are utilized as a source of fuel in rural parts of India. It is estimated that 640 million tons of cow dung is being utilized to meet the house hold fuel requirements. Besides the cow dung, goat extreta, and poultry dropping can also be better utilized for organic manure.

Present Trends:

As a result of various dairy development programmers the country is having presently 233 processing plants and 46 milk products factories. The cooperative public sector plants and organized private plants have an estimated handling capacity of 8.65 million liters per day (MLPD). Various cattle improvement project have been-started in 600 community blocks. The country has now 122 intensive cattle development programmes (ICDP) 140 cattle breeding farms, 40 Exotic cattle farms and 48 frozen semen banks in operation. These activities has resulted in enhancing the milk production by 494.11% in the past three decades although increase in breedable cows and buffaloes 22-23% during the same period. Through a net work of over 4200C milk producers cooperative organized under the operation floot. Programme, a National milch grid has been successfully established. This grid covers besides the four-metropolitan cities. Nearly 200 cities and towns
The fallen and slaughtered cattle and buffaloes also contribute hides and skins, bones and hooves etc. The hides and skins, from cattle and buffalo are estimated at 0.82 million tons annually.

Employment generation:

Animal Husbandry & Dairying may be regarded as a source to create the employment in rural areas all round the year. Indian Agriculture is mainly dependent on monsoon and hence agriculture field faces certain bottlenecks to provide employment during such periods. On an average Agriculture sector may provide 200 days employment to the rural persons. This means they have to find alternate source of employment for income during the rest of the year. Dairy farming, sheep and goat rearing, poultry production, pig farming rabbit rearing are the alternate sources of mix farming. It may be possible to generate the employment for the farmers as well as land less laborers who can do this job themselves, or it may be possible to employ young and the old family persons as a side business. Many of the operations in Animal Husbandry and Poultry Farming can be done by the rural women. It is estimated that on an average 35 million human years/annum employment generation has been potential through this sector.

Terminology Used in Livestock Production

Type: It is a commonly accepted standard that combines those characteristics essential in adopting an animal for a particular purpose e.g. milk, meat wool or work.

Breed: It is groups of animal that are result of breeding & selection have certain distinguishable characteristics.

OR

A group of animals related by decent & which are similar in most of the characters like general appearance, size, colors, horns it is called breed.

OR

A breed may be defined as a cluster domestic animal of a species where individuals are homogenous in certain distinguishable characteristics which differ from one to other group of animals.

Species: A group of individuals which have certain common characteristics that distinguish them from other group of individuals with in species the individuals are fertile when in different species they are not.

Sire: The male parent of the calf.

Dam: Female parent of the calf.

Calf: Young one of cattle or buffalo below the age of six months is called calf.

Heifer: The younger female of cattle above age of six months to first calving.

Cow: The adult female of cattle from the date of first calving is called cow.

Bull: It is unsaturated of, cattle used for breeding or covering the cows.

Bullock: It is the castrated male of cattle used for work.

Service: The process in which mature male covers the female i.e. in heat with the object to deposit spermatozoa in the female genital tract is called service.

Conception: The successful union of male and female gametes & implantation of zygote is known as conception.

Gestation: It is the condition of female when developing foetus in present in the uterus.

Gestation period: The period from the date of service (actual conception) to the date of parturition is termed as parturition period or pregnancy period. This period varies according to species of animals e.g. is cows 279-283 days, in buffalo 310 days, sheep 148-152 days, goat 150-152 days.

Parturition: The act of giving birth to young one is called parturition.

Lactation period: The period after parturition in which the animal produces milk.

Dry period: The period after lactation in which the animal does not produce milk.

Calving interval: The period between two successive calving is calving interval.

Average: It is the sum of production divided by No. of animals.

West average: It is the average daily milk yield of a cow is lactation.

Total milk yield. of a lactation (kg or Lt).

W.A = --
Lactation period (days)

Herd Average: It is average daily milk yield of milling animal in a herd.

$$\text{H. A.} = \frac{\text{Total milk yield of a day}}{\text{No. of milking animals}}$$

Overall average: It is average daily milk yield of the animal in the period of calving interval.

$$\text{O.A.} = \frac{\text{Total milk yield of lactation}}{\text{Calving interval (days)}}$$

Environment: The sum of all external influences to which an individual is exposed.

Genotype: The complete genetic make up of an individual- or its combination of genes it possesses which influences its characters. Several different genotypes may.

Phenotype: The external appearance or some other overall or measurable characteristics of an individual or it is the actual expression of the character as determined by his genes & the environment in which he has lived.

Half sib: Half brothers or half sisters

Full sib: Full brothers or full sister.

Heridity: The occurrence of genetic factors derived from each of its parent in an Individual.

Heritability: The percentage of variation in individual characteristics between related individuals which is due to true genetic difference.

Repeatability: It is the expression of the same trait at different times in the life of the same individual or the tendency of an individual to repeat its performance e.g. dairy cow in successive lactation.

Allel: One or two or more alternative foms of a gene. Alleles are those genes which may appear at same locus in homologus chromosomes.

Gene: It is the unit of inheritance, which is transmitted in gametes or reproductive cells. It is the physical basis of heredity.

Dominance: A gene is said to be dominant when its characteristic effect is expressed in the heterozygote as well as homozygote, i.e. Aa < AA. Ability of gene to cover in block out expression of its allele or genes that have observable effect when present in any one member of a chromosome pair

Recessive: Genes which have no. observable effect unless present in both members of a chromosome pair.

Epistasis: Interaction of two or more pairs of a gene that are not allele to produce a phenotype that they do not produce when they occur separately.

Lethal: (Deadly) A gene or genes that cause death of an individual which are possessed by them during pregnancy or at the time of birth.

Prepotency: The ability of certain individuals to stamp or impress their characters upon their offspring or prepotency is the ability to transmit characteristics to offspring to a marked degree.

Fertility: Ability of an animal to produce large number of living young.

Fecundity: It is the potential capacity of the female to produce functional ova regards of what happens to them after they are produced.

Sterility: Inability to produce any offspring.

Free martin: A sterile heifer born twin with the male.

Cryptorchids: The failure of testes to descend fully into the scrotum. If one testes is in scrotal position the male is usually fertile but if both are retained in the abdominal cavity sterility usually reported.

Atavism: The reappearance of a character after it has not appeared for one or more generation.

Buller: Cow always in estrus condition.

Teaser: A vasectomized (castrated) bull used to detect the heat or estrus of female (cow).

Herd: It is a group of cattle or buffalo.

Flock: It is the group of sheep, goat or poultry birds.

Steer: The male cattle that is castrated when he is still a calf or before the development of sexual maturity is called steer.

Veal: The meat of calf below the age of 3 months.

Beef: The meat of- cattle past calf stage.

Pork: The meat of swine.

Mutton: The meat of sheep & goat.

Chevon: The meat of goat

Wedder: A castrated sheep is called wedder.

Prolificacy: Ability to produce large number of offsprings. The animal is said to be prolific.

Variation: The degree to which individuals differ with respect to the extent of development of expression of characteristics.

Puberty: It is the period when reproductive tract & secondary sex organs/characteristics start to acquire their mature form. Before on set of puberty the reproductive tract of heifer grows proportionately to body growth but beginning at about 6 months age growth rate of these organs is much grater than body growth. At about 10 months of age the rapid growth phase of the reproductive tract ceases & this signifies the end of puberty. Heifer reaches puberty earlier than bull.

Inheritance: Transmission of genetic factors from parent to offspring's.

Germplasm: The material on the basis of heredity taken collectively. The sum of gene constitution of an individual.

Foetus: A term for developing young one during last quarter of pregnancy.

General Information:

Sr. No	Species	Female	Male	Young one	Act of parturition	Average Life Years
1	Cattle	Cow	Bull	Calf	Calving	16-20
2	Local buffalo	Buffalo	Buffalo Bull	Calf	Calving	16-20 .
3	Goat	Doc	Buck	Kid	Kidding	12-15 .
4	Sheep	Ewe	Ram	Lamb	Lambing	12-15 .
5	Swine	Sow	Bore	Litter	Furrowing	8- 10
6	Horse	Mare	Stallion	Foal	Whelping	18-22
7	Ass	Jennet	Jack	Foal	Whelping	14-18
8	Fowl	Hen	Cock	Chick	Hatching	3 - 4
9	Duck	Duck	Drake	Chick (Duckling)	Hatching	4 - 5

Terms Used in Poultry Production

Hen: A matured female chicken generally above 20 weeks of age.

Cock: A matured male chicken above 20 weeks of age.

Pullet: A young female chicken from 9 to 20 weeks of age.

Cockerel: A young male chicken from 5-8 months of age.

Chick: A young male or female fowl below S weeks of age.

Day-old chick: Hatched out chick is called as day-old-chick up to 24 hours.

Grower: A young chick of 9lh week of 20lh week of age of either sex.

Brood: A group of chicks of same age raised in one batch is called as a brood.

Brooding: The process of rearing the young chick from day old stage to 4 to 6 weeks of age during which, heat is to be provided to keep them warm.

Brooder: A device for providing artificial heat to the chicks.

Broiler: They are the hybrid chicks having rapid growth and attaining about 1.5 kg weight during the period of 6 weeks of age. Sold for table purpose within 8 to 10 weeks period. They possess a very tender and delicious meat.

Capon: It is a young male birds of which testicle are removed.

Layer: An egg laying female chicken up to one year after starting the laying of eggs.

Broody: A hen which has stopped laying eggs temporarily.

Clutch: The number of eggs laid by a bird on consecutive days. A clutch of 3-4 eggs is preferred.

Moulting: The process of shading old feathers and growth of new feather in their place moulting normally occurs once in a year.

Culling: Removal of unwanted bird from the flock is known as culling e.g. old non-laying birds, sick birds and masculine hens are removed.

Pause: It is the period between two clutches in which eggs are not laid by hen.

Hen-day-production: This is arrived by dividing total eggs laid in the season by the average number of birds in the house.

Hen-housed-average: This is arrived at by dividing the total number of eggs laid in the season by the number of birds originally placed in the house. No deductions are made for any losses from the flocks.

Animal Management

In India livestock form an integral part of Agriculture. In fact the country's national economy is closely knit with Agriculture and livestock. It is therefore, a matter of great importance that the livestock are maintained in good health and provided proper management "The productive potentialities of livestock are controlled by three principal factors namely i) genetic make up ii) Nutrition iii) Environment including the climatic conditions

What is management?

Management is the art and science of combining ideas, facilities, processes, materials and labors to produce and market a worthwhile product for service successfully.

Manager:

A manager is an organizer and a converter he converts resources in to product. This is just as true for our dairy farm as for our biggest industries; he converts labor soil fertility hay silage and other inputs in to milk. These transformations do not occur by happenstance. They are the results of a purposeful and premeditated force called management.

Function of management:

A manager must perform five major functions. He must plan, organize, coordinate, direct and control. The manager's role is that of a 'decision maker' i.e. who must decide what to do., how to do it, when to do it.

For successful performance of management process, the manager must

1. **Observe**: Gather information about all resources technologies, alternative market outlet Sources of capital credit needed and items affecting successful operation
2. **Establish goals:** Clearly set and the objectives to be achieved.
3. **Identify problems:** Find the obstacles or stumbling blocks which. Hinder the progress to goals.
4. **Analyze**: Compare alternative methods of reaching goals, in terms of capital labor and other assets.

5. **Decide**: Choose a plan of action and set out clear-cut procedure.
6. **Act**: Put chosen plan into operation.
7. **Be responsible:** Assume responsibilities for the consequences, of the actions taken.
8. **Evaluate:** Measure results and compare with goals.
9. **Control:** Keep a careful check on production level.
10. **Adjust:** Keep the operating system flexible to take the advantage of new development tools and management

Tools and management:

Some of the tools which are helpful in acquiring accurate information about dairy business and in guiding management decision are as follows:

1. **Farm records:** No business can be operated successfully without a good system of accounting for a dairy farm. The account should include the following
2. **Complete inventories:** At the beginning and end of the year including a summary of all assets, debts and net worth.
3. Production records (including main animal products - and their byproducts.
4. Current expenditure and receipts-including quantity sold)
5. Annual production and financial summary.
6. Analysis of year's records to determine strong and weak points.

Care and Management of Newly Born Calf

All dairy operations must be planned with due regard to the comfort the animal. After calving the cow will usually be up and will begin to dry the calf, if for some reason the cow is unable to get up then the calf should be dried with a towel or other suitable material.

1. Make sure that all mucus is removed from the nose and mouth. If the calf does not start to breathe, artificial respiration should be used by alternately compressing and relaxing the chest wall with the hands after laying the calf on its side.
2. Naval cord should be cut with sterilized scissors leaving "form the body and them entire naval cord be disinfected by Deeping it into a cup containing tincture of iodine.
3. Normally the calf will be on its feet and ready for suckling the dam within an hour. Some assistance in this stage is useful. Clean the udder before the calf starts sucking.
4. Feed the calf with first milk i.e. colostrum at least for 48 hours. The colostrums should be fed within half an hour after birth. Delay in its feeding causes the calf to loose the ability to absorb antibodies across its inertial walls. The antibodies present in colostrum protect the calf against diseases and it has a laxative effect the rate of feeding should be about 10% of the calf s weight per day up to a maximum of 5-6 liters per day.
5. The colostrum is the first secretion of cow after calving. It is thick and yellow in color. It contains 4 to 5 times more protein and 10 to 15 times more vitamin-A than normal milk. Protein of colostrums contains much higher proportion of globulins. The globulins are to be the source of antibody presumed developing the defense mechanism in the calf for many infections. Colostrum is also rich in minerals like Cu, Fe, Mg andMn. It also contains several other vitamins like Riboflavin, Cholin, Thiamine, Pantothenic acid etc., which are for growth of calf.
6. The calf is best maintained in an individual pen or stall for the first few weeks. After about eight weeks it may be handled with a group.
7. Take body weight of the calf and identify the calf by tattooing.
8. At the age of 15 days 32-40 CC of H.S. serum should be inoculated.
9. Dehorn the calf preferably within 15 days after birth.
10. Teats of the udders of heifers in excess of four should be removed.
11. At the age of 3 months the calf should be vaccinated against Anthrax and fifteen days there after it should be vaccinated against B.Q

The future of any herd depends upon how calves are raised. One has to raise one's own calves to make a good herd. So the calf rearing should be taken upon scientific lines and it should be achieved economically.

Management practice up to six months:

1. Provide fresh, clean water all times, particularly when milk feeding is induced discontinued
2. Giving of identification mark which is necessary for keeping proper records, proper, feeding, better ore and management.
3. Dehorning the calves: at the age of 2-3 weeks, bull calves should be castrated suitably.
4. Castration of bull calf: At age of 2-3 months, bull calves should be castrated suitably.
5. Removal of extra least: In female calves, the following points to be noted
6. Housing: While housing the calves/ the following points to be noted.
7. Calf pen should be close to cow shed.
8. Pen should provide sunlight; good ventilation floor should not be slippery.
9. After 6-8 weeks, calves may be grouped according to age, sex.
10. The feed boxes & watering equipment should be provided in the pen.

System of Calf Rearing

1. Sucking method:

In this method, the calf is allowed to stay with its mother and allowed to suckle only a little before and after of milking the cow. The calf gets whole milk throughout lactation.

Advantages:

i) This is natural system of feeding.
ii) The calf gets contamination free milk.
iii) No much care is required to take during feeding.
iv) The mother-calf affection developed.

Disadvantages:

i) If calf dies, the cow refuses to let the milk.
ii) It can not be ascertained about over feed or under feeding of the calf.
iii) If milk is infected the infection may be to calf.
iv) The actual quantity of milk yield of cow can not be calculated.
v) The post partum heat is late.

2. Weaning method:

In this system, the calf is taken away from its mother either just after the birth or after 2-3 days of birth, sometimes it is allowed till the period of colostrum feeding. After that, the calf rearing is entirely by isolation system.

The immediate step, after weaning of the calf is to teach it to drink milk is very important

1. **Nipple system:** Used for 3-4 days-aged calves. A pail containing milk equipped with rubber nipple used which the calf sucks.
2. **Hand fiddling:** When the calf develops appetite insert two fingers of right hand into the mouth while holding milk in left hand at convenient height for the calf. While calf suckles the fingers, the muzzle is gradually pressed down into milk pan. This way calf learns to drink milk.

Advantage:

i) Cow continues to give milk whether calf is alive or not.
ii) The calf can be culled at an early stage.
iii) It can be fed scientifically as per requirements no problem of under feeding and over feeding.
iv) The actual amount of milk produced by cow can be determined.

v) Milking without calf is more hygienic & sanitary.
vi) Cow becomes regular breeder; the calving interval is less than the unweaned calves.

3. Milk feeding schedule to the calf:

The calf after weaning from the Jam, it should be fed with the whole milk, skim milk and re-constituted milk and also calf starters in gradual age. The temperature of the milk must be body temp. I.e. 39°C, the utensils used must be clean and sterilized; the milk should be fed twice a daily.

Body weight (kg)	Calf age (days)	Colostrums (lire. Per body wt.)	Whole milk (liters per body weight)	Skim milk (liters per body wt.)
Upto25	Upto5	1/10th	-	-
20-30	6 - 20	-	1/ 10th	-
25-50	21-30	-	l/15th	1/20th
30-60	31-60	-	l/20th	l/25th
40-75	61-100	-	l/25th	l/25th

Calf Starters:

It is mixture of grain protein feeds, minerals, vitamins & antibiotics. It has been evolved for use with limited whole milk. An ideal calf starter contains 20% DCP, & 70% TDN.
If the calves raised with calf starter, the schedule is:

Age(day)	Whole Milk (Kg)	Skim Milk (Kg)	Calf Starter in Kgs.
0-5	Colostrum	-	-
6-7	2.75	-	-
8-14	3.25	-	-
15-21	2.75	1.00	0.10
22-28	1.75	2.00	0.20
29-34	1.00	3.00	0.30
35-42	0.50	3.50	0.50
43-56	-	3.50	0.75
57-84	-	2.50	1.00
85-112	-	0.50	1.25
113-140	-	-	1.75
141-182 (up to 6 months)	-	-	2.00

Care of Heifer

Heifers-can be reared by two methods:

i) Outdoor system/grazing method.
ii} Indoor system.

i) Outdoor system:

The heifers are reared mainly on grazing. The following are management points in this system:

1. They should be shifted daily from one grazing land to another.
2. Pasture plots be grazed rotationally containing legume grass.
3. Grazing land must have provision of shade & supply of cool drinking water.
4. Concentrates and minerals may be fed through troughs located in the field.

ii) Indoor system:

In this system, they are confined by compound and provided with shelter. The main points to be considered in this system are;

1. **Feedings:** They should be provided with good quality of hay or roughages & concentrates or grains. The feed must be rich in nutrients like proteins, energy, minerals, and vitamins.
2. **Housing:** The heifers from 6 months onwards should be housed separately from suckling calves and no male calves be kept together beyond 6 months. For better allocation of resting area, calf should be provided with below stated space,

i.e. 20- 5 sq.ft/calf for below 3 months of age
25-30 sq.ft from 3-6 months of age
30-40 sq.ft from 6-12 months of age
40-50 sq.ft from above one year

3. **Exercise:** In this system, the care is to be taken that they should get sufficient exercise which removes stiffness in limbs, -keep thrifty growing & maintain normal appetite.
4. **Culling of heifers:** Those having anatomical defects, bad deposition, poor growth & late maturity should be culled.
5. **Control of parasites:**
   - **De-worming of heifers:** Worms interfere with absorption of food nutrients ultimately interfere with host's growth, therefore heifers be de-wormed after every 4-6 months.
   - **Control of Ectoparasites:** Ectoparasites like ticks, lice etc. should be treated to control such parasites by dipping or spraying with 0.5% BHC or other insecticides like 1% Malathion spray is effective. The regular grooming is also helpful.
6. **Vaccination of heifers:** At 6 months of age, heifer should be vaccinated for & Mouth disease, T.B. & Rinderpest diseases. While older heifer should be vaccinated for Anthrax, Black quarter.
7. **Age of Breeding:** Many factors affect the age of breeding i.e. Breed, system of feeding, and quality of nutrition. Under average manage mental conditions of feeding & care, the heifers attaining weight of 200 kg (minimum) may be considered of age at first breeding.
8. **Steaming up:** A pregnant heifer few days prior to calving must be fed liberally is called steaming up. It is done for the reasons that, heifer continues to grow, she has to bear an unborn viable calf, and she must maintain her good health during lactation period. For steaming up heifers must be given 1.5 kg concentrate mixture.
9. **"Breaking-in" heifers:**
   - **Care in training heifers:** Heifer should be handled with kindness. They should be trained to load with halter from an early age, which helps to make docile cow.
   - **Housing pregnant heifer with milch herd:** This practice to heifers should start about a month prior to Calving to accustom them their place in barn.

Care of Milking Animals

The routine of management practices like feeding and milking and caring should be followed some time each day, being animals are more sensitive habitual for timing.

1. **Feeding & watering:** The adequate clean & fresh water should be provided. An adult dry cow drinks 30-32 liters of water per day besides it requires 4 liters of water for every liter of milk production. Also, the water consumption increases when air temperature rises.

Feeding: The following feed should be fed to cow for one week to recoup energy i.e. 1 kg cooked bajra per day + 1 coconut + 100 gin methi seed + 100 gin shepu + 100 gm Aaliv + 100 gm sweet oil

Regular feeding for milk production: The production ration should. Be given the additional allowance of ration for milk production over and above maintenance requirement. One kg additional amount of concentrates is required for every 2.5 kg of milk.

2. **Housing:** Good housing is required for protecting animals from heat, rains and winds. Also, proper drainage, ventilation and exposure to sunlight must be there. These factors must be available in any type of housing chosen.
3. **Cleaning & grooming:** Cows should be kept clean both for clean milk production and health of animals, it requires daily brushing which removes, dirt and loose hair. The regular grooming helps to keep skin clean, helps for blood circulation.
4. **Disease control:** The prevention of disease & parasite infestation of the herd is most important. To achieve this, keep the sanitation by keeping the housing & other places clean and regularly disinfected. Many diseases are also prevented by timely vaccination.
5. **Exercise:** The cows should be provided free movement to give the needed exercise.
6. **Milking:** The udder and teals should be washed with warm water mixed with $KMnO_4$ solution and wiped to dry before milking solution and wiped to dry before milking. The milking should be conducted cleanly, gently, quietly, quickly and completely by suitable method of milking. It should be completed within optimum time period of seven minutes.
7. **Breeding:** Cow should be bred at 60 days after date of parturition which helps good reproductive health of cow.

Care of Pregnant Animals

The early singe or latter 1/3 period of the gestation period is important period in view of care and management.

- **Feeding:** It is necessary to provide adequate feeding to meet nutritional requirements of both mother and foetus. The challenge feeding (extra feeding) should be given from 5th month of pregnancy @ 1.25 - 1.75 kg of concentrate mixture and give 3.4 - 4.5 kg from 8th month onwards, over and above maintenance ration to Zebu and crossbred animals. Provide adequate clean water.
- **Drying of Cow:** The pregnant cow-should be dried above 60 days before expected dale of calving. To conserve the nutrients which are required for developing foetus & increased milk yield.
- **Housing:** Pregnant animal approaching parturition should be isolated and kept in calving pen which should be clean, well ventilated, bedded and disinfected. This helps to take special Care regarding feeding management, to avoid crowding, mounting by other animals, to avoid infection from oilier animals.
- **Care at expected Date:** To know expected date of calving is a must to take care at time of parturition. Careful watch should be kept close to expected date of parturition. Do not interfere the normal act of calving. If there is dystokia provide time, veterinarian help.

Care of Breeding Bull

The care and proper management of breeding bull is important for success of breeding programme.

1. **Selection:** The breeding bulls should be selected from good pedigree
2. **Feeding:**

- The properly balanced ration should be given which contains adequate energy, protein, minerals & vitamins.
- Feed to male calf after discontinuation of milk, it should be provided with good quality, legume hay and 2 to 2.5 kg of concentrate having 12-15% DCP.
- Feeding to mature bull: Should be fed adequately to keep it on good flesh but not over fat, sufficient amount of green feed, 1 kg of good quality hay (DM) and 1.5 kg of concentrates per 100 kg of body weight per day will keep in good breeding condition.
- The breeding calf if provided with good feeding practices it will develop in a vigorous nature mature bull & reach sexual maturity of young age.

3. **Housing:**
    - The bull should be housed in a separate bull pen measuring 15' X 10' dimension.
    - The stall should open into strongly fenced paddock into which the bull has free access & movement.
    - The pen should have stanchion to which the bull can be tied during cleaning time.
    - The feeding & watering arrangement should be made in the pen and paddock.
4. **Exercise:**
    - It is needed to keep normal appetite, retain breeding power and good health.
    - Males which received plenty of exercise produce larger ejaculation containing more sperms of higher activity.
5. **Training:** Bull should be trained to be lead with bull staff at an early age population is a pressure on limited sources, so timely culling of the unwanted animals is desired.

A. In case of calves the culling is done due to

- Weak birth weight
- Poor type
- Poor growth rate
- Disease infected.

B. The important reasons for culling the cows from the herd are as:

i) Abortion cases ii) Disease infected iii) Sterility iv) Low productivity
v) Udder infection vi) Long dry period vii) Excessive fattening
viii) Old age

C. Reason for culling the breeding bulls is as:

i) Low pedigree ii) Poor semen quality iii) Lack of sexual libido
iv) Disease v) Old age.

The culling of the animals is a continuous process by either of reasons mentioned above and is one of the most .important management practices.

Preparing Animals for Shows and Exhibition

1. Purpose:

a. To exhibit the best type of animal and win the cattle show
b. The individual breeder can exchange ideas & experience in shows.
c. Shows give opportunity for comparison among superior type of animals within and between breeds.
c. It helps new breeders to make contacts with established breeders.
d. Helps in discovering and popularizing better genetic make-up in the various breeds.
e. Increases pride and involvement of the farmers.
f. Helps in uplift of dairy industry.

2. Needed:

i) Best type and true to breed ii) Body brush/comb iii) Soap iv) Rope & rope halter v) Hair clipper/scissors vi) Bucket vii) Warm water viii) Coconut oil ix) Hoot rasp x) Duster/ sand paper.

3. Procedure

1. **Selection of winning type animal:** Select animal of best-type possessing true breed characteristic i.e., dairy conformation, good temperament, proper growth & development, at least before two month of show.
2. **Early preparation of show Animal:**

Cows: Breed the cow on such date it will calve approximately 3-4 days they are exhibited.

Heifers: Young animals of different age groups i.e., upto 6 months, 6 months to 1 year, above 1 year are exhibited on show. In respective age range select best heifer approaching upper age limit for larger size and exhibiting dairy type.

Feeding: Give extra concentrate (Consisting of wheat bran, groundnut/barley and linseed meal) 1 to 1.5 kg per day depending upon condition of the animal. Linseed meal/cake @ 250 gm per day may be fed being improves the condition of coat Feed roughages like legume hay adlib.

Attending horns: If breed posses horns, those should be rasped to make them symmetrical & smooth.

Trimming hooves: Hoof should be carefully trimmed and properly shaped for improving appearance and gait.

Improving condition of cow:

i. Clipping: Head, neck and tail hairs may be clipped; also airs on bell udder should be clipped for distinct appearance.
ii. Brushing: Groom the animal with the help of brush both morning and evening which makes hair coat glossy.
iii. Washing: Wash animal daily with mild soap to keep clean dry body with dusters, tie the animal in the sun for about an hour after washing.
iv. Blanketing: With a piece of coarse cloth or blanket and rub vigorously, on the body to give brighter look to coat
v. Bedding: Spread sufficient clean absorbent bedding in the stall of animal.

Training: Make animal accustomed to the use of halter and leading rope so that it will be easily led in show and animal will stand in correct position to display for judging.

Final preparation on the day of show:

i. Wash animal body & dry it with towel, Rub sopex warm water solution over body with duster, allow it to dry & brush the coat vigorously to give good luster.
ii. Rub the sandpaper over harms & hoofs, apply little coconut oil on horns, hoofs and tail, comb the switch to give clean and fluffy look.
iii. Milking cow should be milked several hours before judging to show better capacity & balanced

quarters.
iv. Keep halters and ropes well cleaned and polished.
v. Animal is given some salt but denied water and offer fresh water before show, such animal will show greater capacity.
vi. Put on clean and tidy dress before entering the show ring.
vii. Be attentive, keep animal calm & at ease.
viii. Display animal in show ring with a pose to display all its best points

Housing for Different Livestock and Poultry

For dairy cattle, care should be taken to provide comfortable accommodation for an individual cattle. No less important is the (1) Proper sanitation, (2) durability, (3) arrangements for the production of clean milk under convenient and economic conditions, etc.

Location of dairy buildings:

The points- which should be considered before the erection of dairy buildings are as follows:

1. **Topography and drainage:** A dairy building should be at a higher elevation than the surrounding ground to offer a good slope for rainfall and drainage for the wastes of the dairy to avoid stagnation within. A leveled area requires less site preparation and thus lesser cost of building. Low lands and depressions and proximity to places of bad odour should be avoided.
2. **Soil type:** Fertile soil should be spared for cultivation. Foundation soil as far as possible should not be too dehydrated or desiccated. Such a soil is susceptible to considerable swelling during rainy season and exhibit numerous cracks and fissures.
3. **Exposure to the sun and protection from wind:** A dairy building should be located to a maximum exposure to the sun in the north and minimum exposure to the sun in the south and protection from prevailing strong wind currents whether hot or cold. Buildings should be placed so that direct sunlight can reach the platforms, gutters and mangers in the cattle shed. As far as possible, the long axis of the dairy barns should be set in the north-south direction to have the maximum benefit of the sun.
4. **Accessibility:** Easy accessibility to the buildings i& always desirable. Situation of a cattle shed by the side of the main road preferably at a distance of about 100 meters should be aimed at.
5. **Durability arid attractiveness:** It is always attractive when the buildings open up to a scenic view and add to the grandeur of the scenery. Along with this, durability of the structure is obviously an important criteria in building a dairy.
6. **Water supply:** Abundant supply of fresh, clean and soft water should be available at a cheap rate.
7. **Surroundings:** Areas infested with wild animals and dacoits should be avoided. Narrow gates, high manager curbs, and loose hinges, protruding nails, smooth finished floor in the areas where the cows move and other such hazards should be eliminated.
8. **Labour:** Honest, economic and regular supply of labour is available.
9. **Marketing:** Dairy buildings should only be in those areas from where the owner can sell his products profitably and regularly. He should be in a position to satisfy the needs of the farm within no time and at a reasonable price.
10. **Electricity:** Electricity is the most important sanitary method of lighting a dairy. Since a modern dairy always handles electric equipments which are also economical, it is desirable to have an adequate supply of electricity.
11. **Facilities, labour, food:** Cattle yards should be so constructed and situated in relation to feed storages, hay stacks, silo and manure pits as to effect the most efficient utilization of labour. Sufficient space per cow and well arranged feeding mangers and resting areas contribute not only to greater milk yield of cows and make the work of the operator easier but also minimizes feed expenses. The relative position of the feed stores should be quite, adjacent to the cattle barn. Noteworthy features of feed stores are given below:

- Feed storages should be located at hand near the centre of the cow barn.
- Milk-house should be located almost at the centre of the barn.

- Centre cross-alley should be well designed with reference to feed storage, the stall area and the milk house of Housing:

Type of Housing:

The most widely prevalent practice in this country is to tie the cows with rope on a Kutcha floor except some" organized dairy farms belonging to Government, co-operatives or Military where proper housing facilities exist It is quite easy to understand that unless cattle are provided with good housing facilities, the animals will move too far in or out of the standing space, defecting all round and even causing trampling and wasting of , feed by stepping into the mangers. The animals will be exposed to extreme weather conditions all leading to bad health and lower production.
Dairy cattle may be successfully housed under a wide variety of conditions, ranging from close confinement to little restrictions except at milking time. However two types of dairy barns are-in general use at the present time.

1. The loose housing barn in combination with some type of milking barn or parlour.
2. The conventional dairy barn.

Loose housing system

Loose housing maybe defined as a system where animals are kept loose except milking and at the time of treatment. The system is most economical. Some features of loose housing system are as follows.

1. Cost of construction is significantly lower than conventional type.
2. It is possible to make further expansion without much change.
3. Facilitate easy detection of animals in heat
4. Animals feel free and therefore, prove more profitable with even minimum grazing.
5. Animals get optimum exercise which is extremely important for better health and production
6. Overall better management can be rendered.

The floor and manger space requirement of dairy cows are give in Table below

Other provisions:

The animal sheds should have proper facilities for milking barns, calf pens/ calving pens and arrangement for store rooms etc. In each shed, there should be arrangement for feeding, manger, drinking area and loafing area.

The shed may be cemented or brick paved, but in any case it should be easy to clean. The floor should be rough, so that animals will not slip. "The drains in the shed should be shallow and preferably covered with removable tiles. The drain should have a gradient of 1" for every 10' length. The roof may be of corrugated cement sheet, asbestos or brick and rafters. Cement concrete roofing are too expensive.

Inside the open unpaved area it is always desirable to plant some good shady trees for excellent protection against direct cold winds in winter and to keep cool in summer.

Table:

Type of Animal	Floor space per animal (Sq. feet)		Manger length per animal (inches)
	Covered Area	Open Area	
Cows	20 – 30	80 – 100	20-24
Buffaloes	25-35	80-100	24 -30
Young stock	15-20	50-60	15 -20
Pregnant Cows	100-120	180 – 200	24 -30
Bulls Pen	120-140	200-250	24 -30

Cattle shed:

The entire shed should be surrounded by a boundary wall of 5' height from three sides and manger etc., on one side. The feeding area should be provided with 2 to 2 % feet of manger space per cow. All along the manger/ there shall be 10" wide water through to provide clean, even, available drinking water. The water trough thus constructed will also minimize the loss of fodders during feeding. Near the manger, under the roofed house 5' wide floor should be paved with bricks having a little slope. Beyond that, there should be open unpaved area (40' X 35') surrounded by 5' walls with one gate. A plan for such a house along with the plan for calves shed and their sections are shown in Fig, 105, it is preferable that animals face north when they are eating fodder under the shade. During cold wind in winter the animals will automatically lie down to have the protection from the walls.

Shed for calves: On one side of the main cattle shed there shall be full covered shed 10' X 15' to accommodate young calves. Such sheds with suitable partitioning, may also serve as calving pen under adverse climatic conditions. Beyond this covered area there should be a 20' X 10' open area having boundary wall so that calves may move there freely.

In this way both cattle and calve sheds will need in all 50 X 50 area for 20 adult cows and followers. If one has limited resources, he can build ordinary, katcha /semi Kutcha boundary walls but feeding and water trough should be cemented ones.

Conventional Dairy Barn

The conventional dairy barn is comparatively costly and is now becoming less popular day by day. However, by this system cattle are more protected from adverse climatic conditions.

The following barns are generally needed for proper housing of different classes of dairy stock on the farm.

1. Cow houses or sheds
2. Calving box
3. Isolation box
4. Sheds for young stocks
5. Bull or bullock sheds.

Cow sheds:

Cow sheds can be arranged in a single row if the numbers of cows are. Small say less than 10 or in a double row if the herd is a large one. Ordinarily, not more than 80 to 100 cows should be placed in one building. In double row housing, the stable should be so arranged that the cows face out (tail to tail system) or face in (head to head system) as preferred.

Advantages of Tail to tail system:

1. Under the average conditions, 125 to 150 man hours of labor are required per cow per year. Study of Time: Time motion studies in dairies showed that 15% of the expended time is spent in front of the cow, and 25% in other parts of the barn and the milk house, and 60% of the time is spent behind the cows. 'Time spent at the back of the cows is 4 times more than, the time spent in front of them.
2. In cleaning and milking the cows, the wide middle alley is of great advantage.
3. Lesser danger of spread of diseases from animal to animal.
4. Cows can always get more fresh air from outside.
5. The head gowala can inspect a greater number of milkmen while milking. This is possible because milkmen will be milking on both sides of the head gowala.
6. Any sort of minor disease or any change in the hind quarters of the animals can be detected quickly and even automatically.

Advantages of face to face system:

1. Cows make a better showing for visitors when heads are together.
2. The cows feel easier to get into their stalls.
3. Sun rays shine in the gutter where they are needed most.
4. Feeding of cows is easier; both rows can be fed without back tracking.
5. It is better for narrow barns.

Floor: The inside floor of the barn should be of some impervious material which, can be easily kept clean and dry and is not slippery. Paving with bricks can also serve ones purpose. Grooved cement concrete floor is still better. The surface of the cow shed should be laid with a gradient of 1" to 1 1/2 from manger to excreta channel. An overall floor space of 65 to 70 sq.ft. Per adult cow should be satisfactory.

Walls: The inside of the walls should have a smooth hard finish of cement, which will not allow any lodgement of dust and moisture. Corners should be round. For plains, dwarf walls about 4 to 5 feet in height and roofs supported by measonry work or iron pillars will be best or more suitable. The open space in between supporting pillars will serve for light and air circulation.

Roof: Roof of the barn may be of asbestos sheet or tiles. Corrugated iron sheets have the disadvantage of making extreme fluctuations in the inside temperature: of the barn in different seasons. However, iron sheets with aluminum painted. tops to reflect sunray bottoms provided with wooden insulated ceilings can also achieve the objectives. A height of 8 feet at the sides and 15 feet at the ridge will be sufficient to give the necessary air space to the cows An adult cow requires at least about 800 cubic feet of air space under topical conditions. To make ventilation more effective continuous ridge ventilation is considered most desirable.

Stall design: The two main types of dairy barn stalls are the stanchion stall and tie stall.

1. The stanchion stall:

It is one of the standard dairy cow stalls. It is equipped with a stanchion for fastening a cow in place.' Usually there is a stall partition in the form of a curved pipe between the stalls to keep the cows in place and to protect their udders and teats from being stepped on by other cows.

The stanchion should be so contracted and arranged as to allow the cows the greatest possible freedom. There should be several links of chain at the top and bottom of the stanchions and sufficient room on each side of it to permit (lie animal to move its head from side to side. It is important to provide for the comfort of the cows and to line them up so that most of the droppings and urine go to the gutter. Practically, it is not possible to fit every cow to her stall properly. To compensate this, many stanchions have adjustments so that they can be set forward if the cow is too large for the stall or backwards if the cow is too small. The cow can be fastened easily and quickly with the stanchions and is held more closely in place than other types of ties. However/ she is held more rigidly and therefore, the stanchion is less comfortable than other types of fasteners.

2. The Tie Stall:

The tie stall requires a few inches longer and wider than the stanchion stall. It is designed to provide greater comfort to the cow. In addition to larger size, the chain tie gives the cow more freedom. Instead of the stanchion, there are two arches, one on each side of the neck of the cow. The cow is fastened by means of rings fitted loosely on the arch pipe,; and connected to a chain which snaps to the neck strap on the cow. The correct space between arches is 10-12 inches. This prevents the cow from moving too far forward in the stall. It is important that in this type of stall, the arches and all other stall parts are kept lower than the height of the cows.

The cow has more freedom in the tie stall then in the stanchion, large cows and those with large udders get along better in them because of freedom they enjoy. It is not desirable to have a tie chain in a small stall.

Manger:

Cement concrete continuous manger with removable partitions is the best from the point of view of durability and cleanliness, A height of l'-4" for a high front manger and 6" to 9" for a low front manger is considered sufficient low front mangers are more comfortable for cattle but high front mangers prevent feed wastage. The height at the back of the manger should be kept at 2'-6" to 3'. An overall width of 2' to 2 W is sufficient for a good manger.

Alley:

The central walk should have a width of 5'-6' exclusive of gutters when cows face out, and 4'-5' when they face in. The feed alleys in case of a face out system should be 4' wide, and the central walk should show a slope of 1" from the centre towards the two gutters naming parallel to each other, thus forming a crown at the centre.

Manure gutter:

The manure gutter should be wide enough to hold all dung without getting blocked, and be easy to clean. Suitable dimensions are 2' width with a cross-fall of 1" away from standing. The gutter should have a gradient of 1" for every 10' length. This wills permit a free flow of liquid excreta.

Doors: The doors of a single range cowshed should be 5' wide with a height of 7', and for double row shed the width should not be less than 8’-9’. All doors of the barn should lie flat against the external wall when fully open.

Calving Boxes:

Allowing cows to calve in the milking cowshed is highly undesirable and objectionable. It leads to in sanitary milk production and spread of disease like contagious abortion in the herd. Special accommodation in the form of loose-boxes enclosed from all sides with a door should be furnished to all parturient cows. It should have an area of about 100 to 150 sq. Ft With ample soft bedding. It should be provided with sufficient ventilation through windows and ridge vent.

Isolation Boxes:

Animals suffering from infectious diseases must be segregated soon from the rest of the herd. Loose boxes of about 150 sq. Ft are very suitable for this purpose. They should be situated at some distance from the other barns. Every isolation box ''should be self contained and should have separate connection to the drainage disposal system.

Sheds for young stocks:

Calves should never be accommodated with adult in the cow shed. The calf house must have provision for daylight ventilation and proper drainage. Damp and ill-drained floors cause respiratory trouble in calves to which they are susceptible. For an efficient management and housing, the young stock should be divided into three groups, viz., young calves aged up to one year, bull calves, i.e., the male calves over one year and the heifers or the female calves above one year. Each group should be sheltered in a separate calf house or calf shed. As far as possible the shed for the young calves should be quite close to the cowshed. Each calf shed should have an open paddock or exercise yard. An area of 100 square feet per head for a stock of 10 calves and an increase of 50 square feet for every additional calf will make a good paddock.

It is useful to classify the calves below one year into three age groups, viz., calves below the age of 3 months, 3-6 months old calves and those over 6 months for a better allocation of the resting area. An overall covered space of:

1. 20-25 square feet per calf below the age of 3 months,
2. 25-30 square feet per calf from the age of 3-6 months,
3. 30-4O square feet per calf from the age of 6-12 months and over, and

4. 40-50 square feet for every calf above one year,

Should be made available for sheltering such calves an air space of 400 to 500 cubic feet per calf is a good provision under our climatic conditions. A suitable interior lay-out of a calf shed will be to arrange the standing space along each side of a 4-feet wide central passage having a shallow gutter along its length on both sides. Provision of water through inside each calf shed and exercise yard should never be neglected.

Bull or Bullock shed:

Safety and ease in handling a comfortable shed for protection from weather and a provision for exercise are the key points while planning accommodation for bulls or bullocks. A bull should never be kept in confinement particularly on hard floors. Such a confinement without adequate exercise leads to overgrowth of the hoofs creating difficulty in mounting and loss in the breeding power of the bull.

A loose box with rough cement concrete floor about 15' by 10' in dimensions having an adequate arrangement of light and ventilation and an entrance 4' in width and 7' in height who make a comfortable housing for a bull. The shed should have a manger and a water trough. If possible, the arrangement should be such that water and feed can be served without actually entering the bull house. The bull should have a free access to an exercise yard provided with a strong fence or a boundary wall of about 2' in height, i.e., too high for the bull to jump over. From the bull yard, the bull should be able to view the other animals of the herd so that it does not feel isolated. The exercise yard should also communicate with a service crate via a swing gate which saves the use of an attendant to bring the bull to the service crate.

Poultry Housing

Poultry is housed for comfort protection, efficient production and convenience of the poultry man.

Essentials of Good Housing:

Comfort: The best egg production is secured from birds that are comfortable and happy. To be comfortable a house must provide adequate accommodation; be reasonably cool in summer, free-from draft and sufficiently warm during the winter provides adequate supply of fresh air and sunshine; and remain always dry. Given these the hen responds excellently.

Protection: Includes safeguards against theft and attack from natural enemies of the birds such as the fox, dog, cat kite, crow, snake, etc. The birds also should be protected against external parasites like ticks, lice and mites.

Convenience: The house should be located at a convenient place, and the equipment so arranged as to allow cleaning and other necessary operations as required.

Location of Poultry House:

In planning a poultry house, the location should be taken into consideration. In selecting site for poultry houses the following factors should be considered.

1. **Relation to other building:** The poultry house should not be close to the home as to create unsanitary conditions. On the other hand it should not be too far away either because this will require more time in going to and for in caring for the birds. In general at least three trips should be made daily to the poultry house in feeding, watering, gathering the eggs, etc.
2. **Exposure:** The poultry house should face south or east in moist localities. A southern exposure permits more sunlight in the house than any of the other possible exposures. An eastern exposure is almost as good as a southern one. Birds prefer morning sunlight to that of the afternoon. The birds are more active in the morning and will spend more time in the sunlight.
3. **Soil and drainage:** If possible the poultry house should be placed on a sloping hillside rather than a hilltop or in the bottom of a valley. A sloping hillside provides good drainage and affords

some protection. The type of soil is important if the birds are to be given a range. A fertile well drained soil is desired. This will be a sandy loam rather than a heavy clay soil. A fertile soil will grow good vegetation which is one of the main reasons for providing range. If the poultry house is located on flat poorly drained soil, the yards should be tiled otherwise the birds should be kept in total confinement.

4. **Shade and Protection:** Shade and protection of the poultry house are just as desirable as for the home. Trees serve as a windbreak in the winter and for shade in the summer. They should be tall, with no low limbs. Low shrubbery is no good as in their presence the soil becomes contaminated under the shrubbery, remains damp/ and sunlight cannot reach it to destroy the di ease germs. One thing we should remember that plenty of sun shines should be available at the site.

Housing requirements:

Floor space: The smaller the house the more square feet are required for each hen. Bigger pens have more actual usable floor space per bird than smaller pens. The recommend at as suggested might be useful regarding floor, feeders and watering space.

For economic production of laying hens it is always better to keep them in small unit of 15 to 25 birds. This number can go up to a maximum limit of 250 birds. In commercial poultry farms units of 125 or so are advisable. Where there is a long house, partitioning at every 20 feet should be made to eliminate drafts, etc.

Ventilation: Ventilation in the poultry house is necessary to provide the birds with fresh air and to carry off moisture. Since the fowl is a small animal with a rapid metabolism its air requirements per unit of body is high in comparison with that of other animals. A hen weighing 2 kg and on full feed, produces about 52 liters of CO_2 every 24 hours. Since CO_2 content of expired air is about 3.5 per cent, total air breathed amounts to 0.5 liter per kg live weight per minute. A house that is a tall enough for the attendant to more around comfortably will supply far more air space than will be required by the bird's that can be accommodated in the given floor space.

Temperature: Hens need a moderate temperature of 50°F to 70°F. Birds need warmer temperature at night, when they are inactive, than during the day. The use of insulation with straw pack or other materials, not only keeps the house. Warmer during the winter months but cooler during the summer months Cross ventilation also aids in keeping the house comfortable during hot weather.

Dryness: Absolute dry conditions inside a poultry house is always ideal condition dampness causes discomfort to the birds and also gives rise to the diseases like colds, pneumonic etc. Dampness in poultry house caused by: (1) moisture rising through the floor; (2) leaky roofs or walls; (3) rain or snow entering through the windows; (4) leaky water containers; (5) exhalation of birds.

Light: Daylight in the house is desirable for the comfort of the birds. They seem more contented on bright sunny days than in dark, cloudy weather. Sunlight in the poultry house is desirable not only because of the destruction of disease germs and for supplying vitamin-D but also because it brightens the house and makes the birds happy. Birds do fairly well when kept under artificial lights.

Sanitations: The worst enemies of the birds, i.e., lice, ticks, fleas and mites are abundant in poultry houses. They not only transmit diseases but also retard growth and laying capacity. The design of the house should be such which admits easy cleaning and spraying. There should be minimum cracks and crevices. Angle irons for the frame and cement asbestos or metal sheets for the roof and walls are ideal construction materials, as they permit effective disinfection of the house. When wood is to be used, every piece should be treated with coaltar, cresol, or similar strong insecticides before being fitted.

Housing Systems of Poultry

There are four systems of housing generally found to follow among the poultry keepers. The type of housing adopted depends to a large extent on the amount of ground and the capital available.

1. Free-range or extensive system
2. Semi-intensive system
3. Folding unit system
4. Intensive system

A. Battery system
B. Deep litter system

Free range system: This method is oldest of all and has been used for centuries by general farmers, where there is no shortage of land.

This system allows great but not unlimited, space to the birds on land where they can find an appreciable amount of food in the form of herbage, seeds and insects, provided they are protected from predatory animals and infectious diseases including parasitic infestation. At present due to advantages of intensive methods the system is almost absolute.

Semi-intensive system: This system is adopted where the amount of free spare available is limited, but it is necessary to allow the birds 20-30 square yards per bird of outside run. Wherever possible, this space should be divided giving a run on either side of the house of 10-15 square yards per bird, thus enabling the birds to move onto fresh ground.

Folding unit System: This system of housing is an innovation of recent years. In portable folding unit's birds being confined to one small run, the position is changed each day, giving them fresh ground and the birds find a considerable proportion of food from the herbage are healthier and harder. For the farmer the beneficial effect of scratching and manuring on the land is another side effect.

The disadvantages are that food and water must be carded out to the birds and eggs brought back and there is some extra labor involved in the regular moving of the fold units. The most convenient folding unit to handle is that which is made for 25 hens. A Floor space of 1 square fool should be allowed for each bird in the house, and 3 square feet in the run, so that a total floor space to whole unit is 4 square feet per bird, as with the intensive system. A suitable measurement for a folding house to take 25 birds is 5 feet wide and 20 feet long, the house being 5' X 5', one third of Ibis fun. The part nearest the house is covered in and the remaining 10' open with wire netting sides and lop.

Intensive system:

In this system the birds are confined to the house entirely, with no access to land outside, and it is usually adopted where land is limited and expensive.

This has only been made possible by admitting the direct rays of the sun on the floor of the house so that par to the windows are removable, or either fold or slide down like windows of railway train to permit the ultraviolet rays to reach the birds. Under the intensive system, Battery (cage system) and deep litter methods are most common.

A. Battery system: This appliance is the inventor's latest contribution to the commercial egg farmer. This is the most intensive type of poultry production and is useful to those with only a small quantity of floor space at their disposal. Nowadays in large cities hardly a poultry lover can spare open lands for rearing birds. For all such people this system will prove worthy of keeping birds al minimum space.

In the battery system each hen is confined to a cage just large enough to permit very limited movement and allow her to stand and sit comfortably. The usual floor space is 14 X 16 inches and the height, 17 inches. The floor is of standard strong galvanized wire set at a slope from back to the front, so that the eggs as they are laid roll out of the cage to a receiving gutter. Underneath is a tray for droppings. Both food and water receptacles are outside the cage. Many small cages can be assembled together; if necessary It may be multistoried. The whole structure should be of metal so that no parasites will be harbored and through disinfection can be carried out as often as required. Provided the batteries of cages are set up in the place which is well ventilated and lighted, is not too hot and is vermin proof and that the food meets all nutritional needs, this system has proved to be remarkably successful in [lie tropical countries. It may be that as it requires a minimum expenditure of energy from the bird, which spends its entire item in the shade, it lessens the load of excess body heat. The performance of each bird can be noted and culling easily carried out. Pullets, which are more often used than birds of over one year, should be placed in the cages at least one month before they are expected to lay.

The feeding of birds in cages has to be carefully considered, as the birds are entirely dependent on the mash for maintenance and production. To supply vitamins A and D, cod liver oil, yeast, dried milk powder are useful/ and fish meal or other animal protein, and balanced minerals and some form of grit must be made available.

As in each cage there will be only pullets so one can never expect fertilized eggs, hence the vegetative eggs will be there, which can be preserved for a longer time than fertilized eggs at ordinary room temperature but can never be used for hatching purposes.

B. Deep litter system: In this system the poultry birds are kept in large pens up to 250 birds each, on floor covered with litters like straw, saw dust or leaves up to depth of 8-12 inches. Deep litter resembles to dry compost. In other words we can define deep litter, as the accumulation of the material used for litter with poultry manure until it reaches a depth of 8 to 12 inches. The build-up has to be carried out correctly to give desired results, which takes very little attention.

Advantages of Deep Litter System:

Safety of Birds: Birds on rage of even in a netted yard can be taken by wild animals, flying birds, etc. When enclosed in deep litter intensive pen which has strong wire netting or expanded metal, the birds and eggs are safe.

Litter as a source of food supply: It may come as surprise to learn that built-up deep litter also supplies some of the food requirements of the birds. They obtain "Animal Protein Factor" from deep litter and some work indicates that this could learn that birds obtain sufficient of this to enable to suitable feed ration to be prepared with only a vegetable protein such as groundnut meal included in the feed. The level of vitamins such as riboflavin increases up to nearly three-fold. According to experiments conducted. The combination of this and the Animal Protein Factor is necessary to good hatchability of eggs and early growth of chickens.

Disease control: Well managed deep litter kept in dry condition with no wet spots around water has a sterilizing action. The level of coccidiosis and worm infestation is much lower watered kept on good deep litter than with birds (or chickens) in bare yards and bare floor sheds particularly where water spillage is allowed.

Labour saving: This is one of the really big features of deep litter usage. Cleaning out poultry pens daily or weekly means quite a lot of work. With correct conditions observed with well managed litter there is no need to clean a pen out for a whole year; the only attention is the regular stirring and adding of some material is needed.

The valuable fertilizer: This is a valuable economic factor with deep litter. According to McArdle and Panda, 35 laying birds can produce in one year about 1 tonne of deep litter fertilizer. The level of nitrogen in fresh manure is about 1%, but on well built-up deep litter it may be around 3 per cent nitrogen (nearly 20% protein). It also contains about. 2 per cent phosphorus and 2 per cent potash, its value is about 3 times that of cattle manure.

Hot weather safeguard: This is an important feature in a hot climate. The litter maintains its own constant temperature, so birds burrow into it when the air temperature is high and thereby cool themselves. Conversely, they can warm themselves in the same way when the weather is very cool. Accordingly, it is a valuable insulating agent.

Animal Health Cover

Herd health programme that emphasize prevention of disease, rather than treatment play a central role in any attempt to increase production efficiency. Treatment will always be important in terms of survival of the individual sick animal. However in terms of survival of the total production unit (profit verses Loss) prevention is the more desirable method of disease control. Individual animal treatment should be viewed as a salvage operation since it occurs after varying amounts of production have already been lost. Under present economic conditions the proverb "an ounce of prevention is worth a pound of cure" is truer than ever before. The selection of drugs and prescription should be left to the discretion of the dairy manager in consultation with his veterinarian.

Health denotes physical, physiological and mental wellbeing of an individual.

Disease means any deviation from normal state of health.

Classification of Diseases:

A. According to mode of origin

1. **Hereditary diseases:** are transmitted from parents to the offspring.
2. **Congenital diseases:** are acquired during intra-uterine life.
3. **Acquired diseases:** are acquired after birth.

B. According to specific causes:

a) Specific diseases: are produced by a specific pathogen or factor. They are subdivided into

i) Infectious diseases: are caused by pathogenic organisms

Viral diseases: Rinderpest (RP), Foot & Mouth disease (HMD)
Bacterial diseases: Black quarter (BQ), Haemorrhagie septicemia (HS)
Protozoan diseases: Surra, Thieleriosis.

ii) Non-infectious diseases: are caused by physical or chemical or Poisonous agents, nutritional deficiency or disturbed metabolism.
E.g.

1. Deficiency diseases - Rickets.
2. Metabolic diseases - Milk fever
3. Poisoning - Pesticide poisoning

b) Non-specific disease: those diseases whose causes are indefinite or multiple e.g. Pneumonia

C. According to mode of spread:

1. **Contagious disease:** sprout by means of direct or indirect contact, e.g. FMD; HS All infections discuses may or may not be contagious but all contagious dieses are injections.
2. **Non-contagious diseases:** do not spread by means of direct or indirect contact. E.g. Rickets.

D According to clinical signs:

1. **Preacute disease** is characterized by very short course (few hours to 48 hours) and very server symptoms e.g. Anthrax,
2. **Acute disease** is characterized by a sudden onset, short course (3-14 days) and severe symptoms e.g. FMD, RP.
3. **Subacutc disease:** whose course is 1-4 weeks and severity is less than acute one. E.g. Sub acute mastitis

4. **Chronic disease:** whose course is more than 4 weeks and signs are not severe in character e.g. Tuberculosis

E. According lo intensity and spread of diseases:

1. **Sporadic disease:** affects one or f-o animals and shows little or no tendency to spread within the herd e.g. Johne's disease.
2. **Enzootic/Endemic disease:** means are outbreak of disease among animals in a definite area or particular district. E.g. Anthrax, H.S.
3. **Epizootic/Epidemic disease:** which assets a large population of animals in large area at the same time and spread with rapidity e.g. FMP, RP.
4. **Panzootic /Pandemic disease:** is a widespread epidemic disease usual of world wide distribution e.g. Influenza
5. **Zoonotic disease:** a disease which can be transmitted from animal to man and vice versa e.g. Anthrax, Brucellosis.

eneral Measures for Prevention of Contagious Diseases

1. Identification of isolation of infected,-rd in contact animal.
2. Treatment of' affected animals
3. Slaughter of animals suffering from incurable diseases.
4. Disposal of Deal animals either burning or deep burial.
5. Destroy contaminant folder by burning.
6. Proper disposal of contaminated water.
7. Regular cleaning and disinfection of cattle shed and its premises.
8. Don’t allow animals from affected to clean area.
9. Restrict the movement of animals from effected to clean area.
10. Don’t allow animals to drink water from ponds, rivers etc. during out break of disease.
11. Close animal markets, cattle shows etc. during outbreak of disease.
12. Regular spraying of insecticide to control external parasites.
13. Regular de-worming to control internal parasites.
14. Avoid stress associated with along distance transportation, inclement weather and under nutrition
15. Provide adequate ventilation and sufficient space.

Table: Normal clinical values in animals:

Species	Temperature 1	Pulse rate, per minute	Respiration rate per minute
Cattle& buffalo Sheep & Goat . Poultry	101.6 102.6 107.0	42 – 60 70 – 80 130 – 160	16 – 24 18 – 30 15 – 30

Hemorrhagic Septicaemia - Animal Disease

Synonyms: Pasturellosis, shipping fever, ghatsurp

It is an actual infectious disease of cattle, buffalo, sheep and goat. It distances transportation. In India, the disease is enzootic in nature. Etiology environmental conditions, malnutrition and long distance transportation. In India, the disease is enzootic in nature.

Etiology: It is caused by *Pasteurella multocida*

Transmission:

1. Ingestion of contaminated feed and water and
2. Inhalation.

Symptoms:

1. High fever (106 - 107°F)
2. Loss of appetite
3. Suspended rumination
4. Dullness and depression
5. Rapid pulse & heart rate
6. Profuse salivation and laciration.
7. Profuse nasal discharge
8. Difficult/snoring respiration
9. Swelling of throat region (submandibular oedema)
10. Death within 10-72 hours.

Diagnosis:

1. History of season, climate & stress factor.
2. Symptoms -high fever, swelling of throat region.
3. Postmortem findings - hemorrhages throughout body & submandibular edema.
4. Examination of blood smears and smears from oedematus fluid.
5. Isolation of the organism from blood & edematous fluid.

Treatment:

Treatment is effective if given in early stage of disease.

a) Specific treatment:

1. Injection. Sulphadimidine @ 150 mg/Kg body weight IV daily for 3 days
2. Injection Oxytetracycline @ 5-10 mg/Kg body weight IV or IM daily for 3 days.

b) Supportive treatment:

1. Use of antipyretics lo reduces body temperature.
2. Use of antihistaminic e.g. Injection Avil/Cadistin5-10 ml IM.

Control:

a) General measures:

1. Isolation and treatment of the affected animals.
2. Close animal markets, cattle shows. Etc.
3. Burning or burial of dead animals.
4. Proper disposal of contaminated feed and water.
5. Disinfection of cattle shed.
6. Avoid long distance transportation and exposure to extreme weather.

b) Vaccination

Alum precipitated M.S. vaccine@ 5 ml subcut every year before monsoon.

Black Quarter - Animal Disease

Synonyms: Black - leg, Farrya

It is an acute infectious and highly fatal, bacterial disease of cattle. Buffaloes, sheep and goats are also affected. Young cattle between 6-24 months of age, in good body condition are mostly affected. It is soil-borne infection which generally occurs during rainy season. In India, the disease is sporadic (1-2 animal) in nature.

Etiology: It is caused by *Clostridium chauvoei*

Transmission:

The disease spreads through

a) Ingestion of contaminated feed and
b) Contamination of wounds.

Symptoms:

1. Fever (106-10S°F)
2. Loss of appetite
3. Depression, dullness
4. Suspended rumination
5. Rapid pulse and heart rates
6. Difficult breathing (dyspnoea)
7. Lameness in affected leg.
8 Crepitation swelling over hip, back & shoulder.
9. Swelling is hot & painful in early stages whereas cold and painless inter.
10. Recumbency (prostration) followed by death within 12-48 hrs.

Diagnosis:

1. History of age, body condition & season.
2. Symptoms - high fever, Crepitation swelling and lameness.
3. P.M. findings - dark colored muscles with gaseous infiltration.
4. Examination of smears made from affected (issues or fluid from 4he swelling.
5. Isolation of the organism.

Treatment:

1. Penicillin @ 10,000 units /Kg body weight 1M & locally daily for 5-6 days.
2. Oxytetracycline in high doses i.e. 5-10 mg/Kg body weight 1M or IV
3. Indcse the swelling and drain off
4. B.Q. antiserum in large does, if available.
5 Injection. Avil / Cadistin @ 5-10 ml IM

Prophylaxis:

a) General measures:

1. Isolation of infected and in contact animals.
2. Disposal of carcass either by deep burial or burning.
3. Proper disinfection of surgical instruments prior to operation.
4. Don't allow grazing in affected area.

b) Vaccination: Alum precipitated B.Q. Vaccine 5 ml subcut each year before rainy season.

Anthrax - Animal Disease

Synonyms: Splenic fever, fanshi, kalpuli

It is an acute widespread infectious disease of all warm blooded animals specially cattle, buffalo, sheep, goat. It is communicable to man i.e. Zoonotic disease. It is soil-borne infection. It usually occurs after major climatic change. The disease is enzootic in India.

Etiology: This disease is caused by bacteria called Bacillus authracis.

Transmission:

1. It usually spreads through ingestion of contaminated feed and water.
2. Sometimes, it also occurs by inhalation and biling flies.

Symptoms:

1. Sudden rise in body temperature (104 - 10S°F)
2. Loss of appetite i.e. off-feed.
3. Severe depression or dullness.
4. Suspended rumination
5. Increased respiration and heart rate
6. Bloat or tympany.
7. Dyspnoea - difficult breathing
8. Dysentery or diarrhoea,
9. Bleeding from natural openings like anus, nostrils, vulva etc.
10. Sudden death in peracute cases.

Diagnosis:

1. History of sudden change in climate and sudden death,
2. Symptom - sudden death & bleeding from natural openings.
3. Postmortem findings:
    1. Oozing of dark: tarry Coloured poorly clotted blood from natural opening
    2. Enlargement of spleen i.e. Splenomegenly.
4. Microscopic examination of blood smears.
5. Isolation and identification of organism.

Treatment : Treatment is effective if given in the initial stage of the disease.

1. Penicillin @ 10000 units/kg body wt. IM.
2. Oxytetracycline © 5-10 mg/kg body wt. IM/IV.
3. Antiantrax serum @ 100-200 ml IV may be given if available.
4. Supportive treatment with antipyretics, antitistammics and fluid thereyp.

Control:

1. General measures: Identification and isolation of affected animals.
2. Movement of animals from infected area to clean area should be stopped.
3. Deep burial of dead animals.
4. Destroy contaminated fodder by burning.
5. Thorough disinfection of cattle shed by using 10% caustic Soda or formalin.
6. Never conduct postmortem of the annual suspected lo be died of Anthrax.
7. Anthrax spore vaccine @ 1 ml subcut every year before onset of monsoon in areas where anthrax outbreaks are common.

Rinder Pest – Animal Disease

Synonyms: Cattle plague, Bovine typhus, Bulkandi

It is an acute highly contagious viral disease of ruminants and pig. Crossbred and young cattle are more susceptible to this virus.

Etiology: It is caused by *paramyxovirus.*

Transmission:

1. It spreads primarily through inhalation.
2. It also spreads through ingestion of contaminated feed and water.

Symptoms:

1. Fever usually p

Foot and Mouth Disease

Foot and mouth disease is a highly communicable disease affecting cloven footed animals and is characterized by fever, formation of vesicles and blisters in the mouth, udders, and teats and on the skin between the toes and above the hooves. Animals recovered from the disease, present a characteristically rough coat and deformation of the hoof. In India, the disease is not serious for livestock and seldom progresses to a fatal issue, but it occurs practically all the year round and, being widespread, it assumes a position of importance in livestock industry. The annual economic loss on account of this disease is estimated to be about Rs. 2.5 crores. The disease affects mostly cattle and pigs of all breeds and ages. Imported cattle and cross-bred suffer more severely, with a mortality rate of 10 to 20 percent, against only 2 to 3 per cent in the local breeds. Although buffaloes, sheep and goats are also susceptible to the disease, they are seldom affected. Under experimental condition, goats have been found to be more susceptible than sheep.

The disease spreads very commonly by direct contact or, indirectly, through infected water, manure, hay and pastures. It is also conveyed by cattle attendants through their clothes or through their hands when the latter have been recently used in milking affected animals. It is known to spread through recovered animals, field rats, porcupines and birds, while improperly sterilized canned meat may also be a vehicle of the infection. Foot-and-mouth disease occurs in a relatively mild form in India, where the disease has been existing for years and where, in view of the frequent opportunities for natural infection, the bulk of the livestock acquire a greater degree of resistance to the disease. It occurs in a virulent form in U.S.A. and United Kingdom and other European countries, where new and susceptible animals are a I lacked a I each mil-break of the disease.

The virus and its characters:

The virus occurs in a high concentration in the lesions of the mouth, feet and udder. At the height of temperature the virus is present in the blood in a low concentration and becomes soon localized in vesicles in the parts of the body mentioned above.

Symptoms:

The virus gains entry into the circulating blood of animal through injury in the lining membrane of the mouth, tongue, intestines, clefts of hooves and other similar parts. The incubation period in natural infection is about two to five days. In artificial infections, the temperature rises to 104 to 105°F. In about 24 to 48 hours and at this stage the virus occurs in the circulation, being eventually carried to distant parts of the body, where it causes the formation of vesicles.

Diagnosis: Its quick spread and the occurrence of lesions in the feet of affected animals are characteristics of the foot-and-mouth disease. It presents some similarity to Rinderpest from which, however, it is readily differentiated by the absence of diarrhoea and by the presence of foot lesions. Confirmation of the diagnosis may, where possible, be obtained by the intradermal inoculation of vesicular fluid or a suspension of vesicular epithelium into guinea pigs which, as a result of such inoculation, usually develop characteristic lesions of the disease.

Treatment:

No scientific evidence is obtainable in support of claims that foot and mouth disease is curable by the use of therapeutic agents. The use of drugs by field workers is only resorted to as a measure of aiding in the natural process of recovery. Thus, the external application of antiseptics contributes lo the healing of the ulcers and wards off attacks by flies. A common and inexpensive dressing for the lesions in the feet is a mixture of coal tar and copper sulphate in the proportion of 5: 1.

Control and prevention:

Prevention is known to be the only dependable method of dealing with foot-and-mouth disease. In countries where the disease does, not exist or where its incidence is very low, legislatie action has been taken to make it obligatory to notify all suspected cases of foot-and-mouth disease. The usual measures adopted in these countries consist in the slaughter of all affected and in-contact animals, a thorough disinfection all utensils and clothes of attendants and a strict watch over animals in the neighboring areas. The slaughtered animals are buried lo a depth to list five feel in the ground and covered with lime and earth. The affected premises are not used for at least 30 days, and are tested for infectivity at the end of this period by allowing small groups of animals into them to commence with.

Enterotoxaemia and Pulpy Kidney – Animal Disease

The first named condition affects adult sheep and the second young lambs, both being caused by *Clostridium welchii,* Type D. Both diseases run an acute course and are highly fatal.

The disease commonly occurs in England, Australia, New Zealand and certain parts of America. Lately, its occurrence has been reported from certain parts of Bombay and Madras. The term 'Pulpy Kidney' denotes a condition in which the kidney is so damaged that it decomposes soon after death, *Clostridium welchii*, Type D, infection in adult sheep usually occurs in those which are in very good condition and over resting is regarded to be a predisposing factor.

The specific toxins liberated by the organisms in the intestines are absorbed very rapidly and death occurs within three to four hours after the onset the disease.

Anti-toxin and whole culture toxoid-vaccine arc very effective for the control of the disease

Mastitis - Animal Disease

Synonyms: Mastitis, Dagadi

Mastitis-denotes an inflanation of the udder, this disease is responsible for heavy financial losses to dairyman due to discarding of abnormal milk, reduced milk production and butter fat, decreased market value of cow and cost of drugs and veterinary services.

In addition to this, the mastitic milk causes dreadly diseases like tuberculosis, brucellosis, sore throat, food poisoning etc. in human beings.

Etiology:

A. Infectious agents:

1. Bacteria - Streptococcus, Staphylococcus, E. coli
2. Viral diseases - Cow pox, FMD
3. Fungus - Aspergillus, Candida, Cryptococcus
4. Mycoplasm

B. Predisposing factors

1. Trail mil or injury to teat and udder.
2. High milk yield.
3. Incomplete or irregular milking.
4. Improper milking techniques.
5. Pendulous udder and long cylindrical teats.
6. Rough flooring.
7. Unhygienic conditions.

Transmission: It spreads through infected water, contaminated bedding, utensils, milkers hands.

Symptoms:

a) Acute form:

1. Fever
2. Loss of appetite.
3. Udder is swollen, hot and painful.
4. Milk may be yellowish or brownish,
5. Milk contains flakes or clots.

b) Chronic form:

1. No swelling of udder.
2. Udder becomes hard due to fibrosis.
3. Milk may show visible changes on careful examination.

Diagnosis:

1. Physical examination of udder i.e. shape, size and consultancy
2. Strip cup test - A strip cup consists of a flat black enamelled plate partitioned into four areas. The milk form all four quarters is stripped directly into cup. The presence of clots or flakes will indicate abnormality of milk.
3. California Mastitis Test (CMT) - This test requires plastic paddle with four chambers. Milk is stripped directly into chambers. The CMP reagent is added in the equal quantity. The milk and reagent is rotated by movement of the paddle and the reaction is observed immediately. Formation of greenish blue precipitate or jelly like clot indicates positive test.
4. Isolation of the organism from milk.

Treatment:

1. Evacuate the Under
2. Intramammary antibiotic therapy/infusion with Vetclox plus or Pendistrin - SH or Tilox @ I tube twice daily for 3 days.
3. Milk should not be used for human consumption forecast 72 hours after last infusion.
4. Injections of antibiotics like penicillin, streptomycin, ampicillin, tetracycline or chloramphenicol IM.
5. Hot fomentation of udder with magnesium sulphate to relieve inflammation.

Control:

1. Isolation and treatment of affected animals.
2. Treatment of all teats of all cows at drying.
3. The healthy non infected cows should be milked first and known infected cow should be milked at last.
4. The udder of cow and hands of milker should be washed with antiseptic solution before and after milking.
5. The floor of the milking shed should be washed with running water.
6. Cows should be provided with soft bedding following parturition.
7. Unsterile objects should not be passed in teat.
8. Teat sores should not be neglected and treated at an earliest.
9. Regular testing of cow milk for mastitis.
10. Use of proper milking method i.e. full hand milking followed by stripping.
11. Protect teats and udder from injuries.
12. Maintain hygienic conditions in cattle shed.
13. The non-responsive quarter should be permanently dried up.
14. Culling of non-responsive cases.
15. Proper disposal of mastitis milk.

Raniket Disease (New-Castle Disease)

It is a widespread, highly contagious infection of the respiration and nervous systems of nervous system of poultry. Mortality due to the Raniket disease may be as high as 100 per cent in young flock. It may also affect the laying birds where mortality is not so high. It affects chickens mostly, but sometimes it also affects turkeys and other fowls. It can also affect man showing localized eye infection.

Casual organism:

Many strains of viruses producing this disease have been Isolated.

Transmission:

Directly from bird to bird through nasal or mouth discharges, by air, or by contaminated feed and litter.

Symptoms:

Respiratory difficulty with a pronounced gasping, coughing and rattling of the windpipes, the nervous sings usually appear 1 or 2 days after the respiratory signs occur. These may consist of partial or complete paralysis of one or both legs, tremor of head and in coordination of neck muscles. They follow a rather characteristic attitude- the head down between the legs or straight back between the shoulders.

On post-mortem examination there is hardly any lesion lo differentiate Raniket from other respiratory disease. However in most of the hemorrhages of proventriculus and hemorrhagic ulcers in intestine can be seen. Laboratory tests, along with certain signs are best diagnostic aids. When diseased bird posted thickening of the air sacks and mucus in the trachea or wind pipe is seen. In adult’s flocks, egg production drops in two or throe days lo almost zero.

Prevention:

Vaccinate the birds against the Raniket disease. It is safe and effective.

Control:

There is no effective treatment or control of the disease. These days it has been seen that Raniket attacks mostly old laying flocks duo 16 decrease in tin - blood antibody levels. Therefore, laying flocks can be revaccinated with water soluble vaccines to boost up the antibody level without causing any stress.

Marek's Disease - Poultry Disease

Marek's disease (MD) till 1963-64 was considered under Avian leucosis complex. As in many other countries, in India also this disease has reached at a dangerous level resulting in yearly loss of over Rs. four crores through mortality. Considering economic loss duo lo morbidity in addition to morbidity it can be well visualized.

Transmission:

It is not transmitted trough egg but it is highly contagious and spreads principally by way of air saliva, nasal washing, feather follicle and faeces from infected birds.

Symptoms:

The disease affects the birds in the age group 6 to 10 weeks and may continue even in laying bird though less in .severity. II generally occurs in two forms: Acute forms-cause sudden death of the affected birds with present of visceral tumors inside the body.

Classical form:

Characterized by bilateral or unilateral paralysis of legs, wings or neck due to enlargement of various nerves inside the body, besides these two typical forms, there are other manifestation of the disease such as blindness and skin tumors.

Prevention:

Immunization has been found to be effective. Recently Herpes Turkey Virus (HTV) vaccine manufacturers in India have been introduced for field trials. There are following immunization procedures:

1. Turkey's help virus.
2. Attenuated virulent MD virus strains.
3. Naturally occurring mild strains.
4. Chicken's blood containing mild MD agent but free from other agents. HTV vaccine is most effective and relatively free of problems

Control:

Three approaches are being practiced to control Marek's disease

1. Breeding stock resistance to M.D.
2. Maintaining; birds under strict isolation,
3. Usa of preventive vaccines.

Chronic Respiratory Disease (CRD) - Poultry Disease

The disease has been reported in chickens and turkeys. CRD is specific disease caused by one of the group of organisms known is pleuro pneumonia like organism (PPLO), but more closely defined is Mycoplasma; the particular organism directly associated with CRD is *Alycoplasma gallisepticum* with or without any secondary complications. According to I he recommendation of FAQ committee meeting held in May l969, the term "Avian Respiratory Mycoplasmosis" (ARM) be used in uncomplicated outbreaks involving only pathogenic avian PPLC (Mycoplasma) and the term CRD be used when PPLO infection is superimposed with other condition in I eel ion is superimposed with other condition.

The mortality entirely to CRD is negligible, but it is important because it predisposes the birds to infection for other disease producing organisms.

Transmission:

M. gallisepticum is transmitted through eggs but organisms can also pass from bird to bird through nasal discharges and through droppings. It can also be transmitted by hands, feet and clothes of attendants of visitors. Symptoms: Uncomplicated CRD is frequently sub-clinical. When symptoms are present they are normally milk in nature and include coughing, sneezing, and a nasal discharge. In turkeys, sinuses are frequently swollen. On postmortem examination the trachea may be found inflamed and the air sacs thickened with pus. The condition affecting air sac is often referred as "air sac" disease but it is more pronounced when other factors including bacteria, complicate the original CRD infection.

The- organism has long incubation period of 10 to 30 days. Therefore only few outbreaks-seen in birds under 4 weeks old

Prevention and control:

As uncomplicated CRD is not a major problem today, it may be necessary to protect only against the probable complicating factors. Increased ventilation without drafts reduces the spread and severity of CRD. Buying replacement stock from CRD free source greatly reduces the risk of spread. A very high standard of hygienic condition is of course, supremely important.

Treatment:

The tetracycline group of drugs is useful in treatment, if given continuously for over a week, as soon as disease's seen in flock, at the rate of 100-400 g per ton of feed. It can also be given through water. Nitrofurans especially furazalidone is very effective. Streptomycin may be injected in sinuses after removal of mucous by spraying in turkeys are helpful.

Coccidiosis: Bird Disease

Coccidiosis is a disease caused by the invasion of the digestive tract by microscopic single celled protozoan parasites, called Coccidia. Coccidia are almost ubiquitous parasites and rearing birds away from them is practically impossible. A mild coccidiosis infection, kept under control, is not very harmful and is actually necessary for creating immunity in replacement flocks. However, a severe attack of coccidiosis can cause great loss to poultry fanners due to mortality and morbidity. There are hundreds of types of Coccidia, but only few (of genuses Eimeria) ore important to poultry farmers. These include Elmeria, Lenella, E. necatrix, E. maxima, E. burnelli.

Transmission:

Coccidiosis spreads from bird to bird through eating or drinking contaminated food, water, litter or other material contaminated with Coccidia. Oocysts are carried mechanically by attendants, equipments, animals, wild birds or insects from place to place.

Symptoms:

Most of the external symptoms like ruffled feathers, unthrift ness paleness, loss of appetite and blood in droppings are the result of destruction of inner lining of intestine including cecae which hinders in the absorption of feed materials from gut to blood.

Prevention and control:

It is very difficulty to avoid contact of bird with Coccidia, hence the sound management practices should be followed to allow the birds to build up immunity and yet keep the disease under control. Feeding of law level of a good coccidiostat though has been found quite effective. Treatment: Several drugs are used in feed to treat coccidiosis may also be given in water as shown in table.

Drugs	Level	Mod of Administration
Sodium Sulphadimidine (Sulmet of Canamide)	0.2 % in drinking water	3-2-3 interrupted schedule (i.e. medicated water for 3 days, no drugs for 2 days and again medicated water for 3 days
Sodium Sulphaquinoxaline (Embazin of May and Baker)	0.05 % in drinking water	3-2-3 interrupted schedule for E. Tenella
Nitrofurazone soluble (Bifuran of Smith Kline and French)	0.011% in drinking water	3-2-3-2-3 interrupted schedule fro E. necatrix 7 days continuous
Amprosol (Merek)	30 g in 20 liter water	6 days continuous
Sulphadimethoxine (Agribon of Roche)	0.05 % in drinking water	7 days continuous

Infectious Bursal Disease (IBD-GUMBORO)

Gumboro is very important viral disease noted in almost all part of the world and was first reported in 1962 in U.S.A. Sudden onset in young chicks preferably broilers in acute form with high mortality is a characteristic symptom. Mortality reaches at peak quickly and returns to normal usually after a course of S to 10 days. The survivals show retarded growth and often unproductive.

The affected chicks exhibit off feeding, huddling, depression, diarrhoea, ruffled feathers, trembling and staggering gait.

On postmortem severe inflammation of bursa of fabricious with enlargement and liquefaction is observed which a characteristic lesion is. Similarly, hemorrhages may be noted in buvsa and body muscles with swelling of kidneys showing poleness and urate deposition may not encounter high mortality but after effects arc severe. The retarded growth has already been discussed along with it profuse damage of bursa leads to disturbance to whole immune system is noted. This results in lowering down of resistance to other diseases also leading to mortality due to secondary invaders.

Prevention and control:

In our country in middle 1990's i.e. 1995-96, more virulent from of IBD has caused severe-up to 45-50 per cent mortality in broilers and cockerels also. The mortality-was very high in thickly populated areas of Andhra Pradesh. Control of IBD involves three important steps. First is to observe scrupulous sanitation and hygienic precautions.

The virus is highly sustainable; hence it is quite possible it will not be completely destroyed by disinfection and cleaning. Therefore, sufficient rest to sheds plays important role in control measures. Secondly, protection by live virus vaccines by eye drops or through drinking water correctly is important Sometimes it may be needed to vaccinate birds 2-3 times also.

Thirdly, care should be taken at breeder level by using inactivated or killed vaccines against Gumboro to pass on effective parental immunity to chicks to protect them for initial 3 to 4 weeks. If maternal immunity is broken very early due to more virulent form, the birds may have to be protected from day-old also.

Udder Structure and Physiology of Milk Secretion

Structure of udder: The udder is located outside the body wall and it attached to it by means of its skin and connective issue supports.

The secretary portion of the udder consists of countless alveoli or chambers lined with individual cells. Each of these alveoli is drained by a small duct which leads to larger ducts. Clusters of alveoli resembling a bunch of grapes are drained by ducts of increasing size until some 10 to 20 ducts conduct milk into the gland cistern. The gland cisterns continue into the teat sinus or cistern. At i he tip of (he teal there is a sphincter tightly closing the outlet of the teal sinus.

Each alveolus is-supplied blood through tiny capillaries which lie outside the secretary cells. Small muscle fibers also surround each alveolus and are important in the removal of milk from the gland. The individual secretary cell is the primary factor in milk production. It extracts all of the components of milk from (he blood stream and either arranges them into new compounds or passes them through directly into the alveolus.

The milk Leldown mechanism:

When milk secretion has continued for a considerable time after milking, the alveoli, ducts and gland and teat cisterns are filled with milk. Milk in the cisterns and larger ducts can be removed readily. Milk in the smaller duels and alveoli does not flow out easily. However, the cow and other mammals have developed a mechanism for releasing milk from the mammary gland. Stimulation of the central nervous system by something associated with the milking process is necessary to initiate the read ion. Stimulation of nerve endings in the teats that are sensitive to touch, pressure, or warmth is the usual mechanism. The suckling action of the calf is ideal for this. However, massaging the udder or washing with warm water is also equally effective. Stimulation is carried by the nerves to the brain which is connected. With the pituitary gland located its base. Mechanisms are activated in the pituitary gland which causes the liberation of a hormone oxytocin from its posterior lobe. Oxytocin is carried by the blood stream lo the udder where it acts on the small muscle rolls surrounding the alveoli, causing them to contract. The pressure thus created forces the milk out of the alveoli and smaller ducts as fast .is it can be removed from the teat.

The letting down process can be stimulated within half to one minute's time. The effective time of the hormone is limited and milking should be completed within seven minutes if all the milk is to be obtained.

Clean Milk Production

Pre-requisites for Good Milking:

Milking is the key operation on a dairy farm; it depends on the income derived. Any amount of scientific feeding or possession of high yielding cows will not help if the milking is inefficient.

Milking is an art requiring experience and skill. Milking should be conducted gently, quietly, quickly, cleanly and completely. Cows remaining comfortable yield more milk than a roughly handled and excited cow. Maintenance of clean condition in the milking barn results both in better udder health and production of milk that remains wholesome for longer time. The act of milking should be finished within 5 to 7 minutes, so that the udder can be emptied completely so long as the effect of oxytocin is available. Complete milking has to be done, lest the residual milk may act an inducer for mastitis causing organisms and the overall yield may also be less.

Preparation for Milking:

The milking barn should be thoroughly washed and scrubbed after each milking so that the barn will be clean and dry, before the subsequent milking is commenced. No dusty feed should be fed during milking. The hind quarters and thighs of cows should be brushed, and washed if lot of filth-is accumulating on them. Buffaloes should invariably be washed during summer; during severe winter brushing should be resorted to. Just before milking (after suckling by calf, if weaning is not practiced) the udder should be wiped with a cloth dipped and squeezed in some weak antiseptic solution. In winter the cloth may be dipped in warm antiseptic solution.

A part from cleanliness of cows and their udders, the milkers as well as the milking pails should be clean. The milkers should wear clean dress and cover (heir heads with suitable caps, lest loose hairs may fall in milk. Their nails should be well trimmed and their hands clean and disinfected between each milking by washing in antiseptic solution. Milkers obviously ill and having filthy habits like spilling, blowing nose etc. should not be used.

After each milking the milking pails should first be washed with warm water, scrubbed well using suitable dairy sanitizer and then rinsed well with clean cold water. Afterwards, they should be stacked neatly in racks -upside down, until next milking. Milking cans should also be treated similarly. Sanitary milking pails with dome-shaped top should be used instead of open buckets or vessels. A milk strainer should invariably be used before milk of each animal is poured into the milking can.

Pay attention to the routine of milking operations. Milch animals are sensitive animals. They get accustomed to certain routines and any sudden change in the routine will disturb them resulting in reduced yield. Experienced milkers should be put on first calver cows while/novices should first be tried on older cows. An ideal proposition is to rotate milkers among a group of cows so that the cows will get accustomed to all. Also milk cows at the same home every day. Any change in timing of milking or even change in ration should be brought about gradually.

Milking Procedure:

In India hand milking of cows is still the most common practice. Cow's arc milked from left side. The order of milking the various teals also differs. Tents may be milked cross wise or for equareters together and then hind quarters together or teats appearing most distended milked first. The milk must be squeezed and not dragged out of teats. The first few strips of milk from each teat should be let on to a strip cup to see clues in milk for possible incidence of mastitis. This also helps in getting rid of bacteria which have gained access and collected in the teat canal.

Stripping and full-hand milking are the two commonly used methods of milking. Stripping consists of firmly seizing the teat at its base between the thumb and forefinger and drawing them down the entire length of the teat pressing it simultaneously to cause the milk to flow down in a stream. The process is repealed in quick succession. Both hands may be used, each holding different teat, stripping alternately.

The full hand method comprises of holding the whole teat in the first finger encircling the teat. The base of the teat is closed in the ring formed by the thumb and forefinger so that milk trapped in the lent sinus may not slip bad-, into the gland eastern. .Simultaneously, teal is squeezed between the middle, ring and little fingers and the hollow of palm, thus, forcing the milk out. This process should be repeated in quick succession. By maintaining a quick succession of alternate compressions and relaxations the alternate streams of milk from the two teats sound like one continuous stream. Many milkers tend to bend their thumb in, against the teat while milking. This practice should be-avoided as it injures the teat tissues.

Full hand milking removes milk quicker than stripping, because of no loss of time in changing the position of the hand, Cows with large teats and she-buffaloes are milked by full-hand method; but stripping has to be adopted for cows with smaller teats for obvious reasons, Full-hand method is superior to stripping as it simulates the natural suckling process by calf. Stripping causes more irritation to teats due to repeated sliding of fingers on teats; and so discomfort to cows. In spite of these drawbacks when all milk that is available is drawn out by full-hand method, stripping should be resorted to with a view to milk the animal completely; the last drawn milk is called stripping and is richer in fat.

In India, milkers are mostly accustomed to meet hand milking. They moisten their figures with milk, water or even saliva, while milking. This should be avoided for the sake of cleanliness. Wet-hand milking makes the teats look harsh and dry chafes, cracks and sores appear which are painful to animal. The hands should be perfectly dry while milking.

When cracks and sores are noticed on teats, some antiseptic ointment or cream should be smeared over them after milking.

Male Reproductive System

The primary, secondary and accessory sex organs are collectively termed as reproductive system i.e. Primary sex organs - testis. Secondary sex organs - Vasa efferentia epididymis, the vasa deferentia & penis. Accessory sex organs - prostate gland, seminal vesicles, bulbourethral glands (Cowper's glands).

The Testis:

Anatomy: Two in numbers suspended vertically within sac known as scrotum, ovoid in shape. Length is 10 - 16 cm & 8 Cm width.

Histology: Each testis composed of several crypts enclosed in serous layer called tunica vaginalis. Each crypt has several numbers of seminiferous tubules. The wall of seminiferous tubules consists of basement membrane & multilayered sperm producing epithelium having two types of cells i.e.
(i) Germ cells -spermatozoa produced
(ii) Sertoli Cells - sperms get matured. The space between seminiferous tubules occupied by interstitial cells (Leydig's cells) produces male hormone.

Epididymis:

Anatomy: is considered in three parts i.e.
(i) Caput (head),
(ii) Corpus (body),
(iii) Cauda (tail).

It arises from efferent ducts testis. Throughout of its length epididymal tube is lined with secretary cells.

Histology:

In caput, tube is lined with ciliated pseudo stratified epithelium, the flagella of which whip in direction of efferent flow.

Physiology:

Spermatozoa produced in testis accumulate & mature during their journey through epididymis which is 30-35 meters in bull.

Transport:

Sperms transported from rete testis to efferent duct by the fluid pressure of testis & by active beating of cilia. It lakes 7-9 days for any sperm to travel from germinal epithelium to cauda

Concentration:

Dilute sperm concentration originated in testis- water absorbed into epithelial cell of epididymis mainly in caput & highly concentrated sperm left in cauda (tail)

Maturation:

In the course of migration of sperm cells get matured as; it result of secretion from epididymal cells.

Storage:

Cauda (tail) is .store depot for sperms where (hey remain viable up to 60 days

Vas deferens:

Anatomy:

It is slender tube with thick cord like wall originating from tail of epididymis ending into urethra. It is paired and is with spermatic arteries, veins, nerves. It passes through the inguinal ring and pelvic cavity.

Histology: It is abundantly supplied with nerves & by voluntary contractions of musculature/ it is involved in ejaculation.

Uretha:

It is common passage way for product of testes, accessory glands & for excretion of urine. It extends through penis to the glands penis.

Penis:

It is male organ of copulation arid composed of erectile tissue attached and held by sigmoid flexure. It has function of ejaculation & excretion of urine.

The accessory sex organs mainly provides bulk of seminal plasma which is rich in carbohydrates, salt of citric acid, proteins, amino acids, enzymes, vitamins which are secretions of accessory glands, i.e. Seminal vesicles - two in number located on either side of ampulla the secretion contains mainly fructose & citric acid contributes to seminal plasma.

Prostage Glands:

Consist two joined parts. It is surrounded by urethral muscles. Secretion is high in mineral content.

Cowper's gland:

Are paired, round - compact of walnut size, located above urethra. Secretion is viscid & mucus like.

Female Reproductive System

It consists of organs, namely

1. Ovaries : Reproductive glands
2. Fallopian Tubes : Coveys ova from ovary to uterus.
3. Uterus : In which fertilized ovum develops.
4. Vagina : dilatable passage from uterus to Vulva.
5. Vulva : Terminal segment of system

Ovaries:

Anatomy:
Two in number laying in the abdominal cavity sizes are 0.5 to 1.5 Inch diameter and 0.5 to 1.5 inch width & thickness.

Function:
Dual purpose - production of eggs or ova and production of female hormone i.e. estrogen

Oviduct (Fallopian Tube):

Anatomy:
Are slender, zigzag lubes attached to ligament 20-25 cm in length, close to ovaries in such a way that eggs / ova released by ovary area cached through funnel shape wide end called as "Infundibulum".

Function:
The epithelial lining of oviduct is cliated of which ciliary motion helps to conduct ova from ovaries to uterus. The fertilization occurs in the ampullary region.

Uterus:

Anatomy:
It consists of short medium body, pair of spirally twisted internally cavity connecting two horns known as body of uterus. The uterus has three layers i.e. outer servosa, middle muscular is & inner mucosa. In non-pregnancy period uterus lies in the pelvic cavity which descends into abdomen during pregnancy.

Function:
Fertilized ovus /embryo develop into uterus until the time of birth. To nourish the developing foetus through cotyledons of inner layer

Cervix:

Anatomy:
It is thick walled portion which lies between uterus and vagina having muscle layers forming longitudinal folds forming spiral passage way through it. It is 4 inch long & 1 inch or more thick.

Function:
It is tightly closed during pregnancy and anoestrus period and refoxen during estsus and parturition.

Vagina:

Anatomy:
It is between cervix to vulva in cow. It is 8-10 inch long. Highly elastic organ.

Functions:
Responsible for secretion of mucus, serves as birth canal dung parturition & admits male organ during copulation.

Vulva:

Anatomy:
It is external vertical opening of genital tract just below anus. Diameter is larger than that of vagina.

Function:
Vulva walls supplied with glands which are active during excitement,

Spermatogenesis

Definition:

The process by which sperms are formed within the seminiferous tubules from spermatogonia or sperm mother cells which lie on the basement membrane

Spermatogenesis is divided in to two distinct phases :

1. Spermatocytogenesis: A series of divisions during which spermatogonia form spermatids.
2. Spermatogenesis: A phase where spermatids undergo metamorphosis forming spermatozoa. The entire process takes about 60 days in bull and 49 days in ram.

Spermatocytogenesis:

1. **Phase – 1:** (15 to 17 days duration) the first step in Spermatocytogenesis is mitotic division of spermatogonium. The dormant spermatogonium remains in the germinal epithelium near the basement membrane to repeat process later on. The active spermatogonium will under go 4 mitotic divisions eventually forming 16 primary sperrnatocytes.
2. **Phase -2:** (15 days duration): Mitotic division of primary spermatocytes during which the number of chromosomes is halved (meiosis-I).
3. **Phase - 3 (few hours):** Division of secondary spermatocytes in to spermatids (meiosis Vll).
4. **Phase - 4:** The 4 spermatids from each primary spermatocyte or 64 from each active spermatogonium.

Spermatogenesis:

During the metamorphosis, the nuclear material compacts in one part of cell forming the head of spermatozoa, while the rest of cells elongate forming the tail. The acrosome a cap around the head of the spermatozoan will be formed. The golgi apparatus of spermatids, the cytoplasm from the spermatids is cast off during formation of tail. A cytoplasmic droplet will form the neck of spermatozoa. Newly formed spermatozoa will be released from sertoli cells and forced out through the lumen of seminiferous tubules in to Retetestis.

Oogenesis (Ovigenesis)

Oogenesis:

It is the formation and maturation of female gamete. Oogenesis begins in the pre-natal period. In the female foetus, the germinal epithelium forms in to clusters in which one genocyte differentiates in to an Oogenium containing typical cell constituents. The Oogonia then undergo proliferation prior to or shortly after bird resulting in the fikled ovaries containing the sole reservoir of all future ova called Oocytes.

The growth of Oocytes is characterized by
i) The enlargement of cytoplasm by accumulation of different sizes of granules deutoplasm (yolk)
ii) The development of an egg membrane - zona pellucida. iii) The mitotic proliferation of follicular epithelium adjacent tissue.

There are two stages in the growth of Oocytes:

First Phase: the growth is rapid and intimately associated with the development of ovarian follicle. Attainment to its mature size occurs at about the time of anstrum, formation begins in the follicle.

Second Phase: The Oocytes doesn't grow in size while the ovarian follicle responding to pituitary hormone increases very rapidly in diameter. During the later phases of follicular growth the Oocytes undergo maturation.

The nucleus which has entered in the prophase of the meotic division during the growth of Oocytes prepares to undergo reduction division. The nuclei and nuclear membrane disappear and. chromosome coalesces in compact form. The centrosome then divides into two centrioles around which aster appear. These asters become more apart and spindle is formed between them. The chromosome in diploid pairs are set free in cytoplasm and become arranged on the equatorial plate of spindle (metaphase-I).

The primary Oocytes now undergo two meotic divisions. In the first division two daughter cells arise each containing one half of chromosome complement. However, one cell acquires almost all the cytoplasm. This cell is also known as secondary Oocytes. The other much smaller cell is known as first polar body. At second maturation division, the secondary oocyte divides into ootids (n) and second pollar body (n).

The two pollar bodies containing very little cytoplasm are entrapped in the zona pellucida of the oocytes and there they degenerate. The first polar body may also divide. The zona pellucida may contain one, two or three polar bodies.

It should be pointed out that it is the secondary oocyte which is liberated at ovilation (primary oocyte in case of horse). The oocyte continues the process of maturation until fertilization when it becomes zygote. In the process of Oogenesis primary oocyte gives rise to one egg.

Oestrus Cycle

Definition:

1. The interval from the first signs of sexual receptivity at Oestrus (heat) to the next estrus is called estrus cycle.
2. The chain of physiological events that begins at one Oestrus period and ends at next is called as Oestrus cycle.

The cycle is of 20 + 2 days in cows for normal female in quite regular cycles. This cycle may be studied in four distinct phases as designated by Marshall i.e. Proestrum, Oestrum, Metestrum and Dioestrum.

Proestrum: (Pre-estrus)

This phase is indiction of animal coming in heat. The ovary is surrounded by follicular fluid containing high level of estradiol. The Graafian follicle within ovary grows. The increased level of estradiol is absorbed into blood making effect to oviduct causing growth of cells lining the tube & increasing in the number of cilia which are shortly helpful to transport ova to uterus. Also, epithelial wall of vagina increases in thickness to accommodate smooth coitus this period is of S hrs to 2 days.

Oestrum: (Estrus)

This is period of sexual desire. The Graffian follicles are mature or ripe in this stage. This phase period comes to close by rupture of follicle of ovulation i.e. release of Ovum. This period lasts for 12-24 hours in cow while 1-2 days in ewe. The symptoms exhibited during this period by cow are it bellows frequently, mounts other animals, excited, licking to other animals and stands to be ridden by other animals. This period is called period of standing heat. The proper period to breed is 8 to IS hrs, for getting high fertility rate.

Metostrum: (Meta-estrus)

Period during which reproductive organs return to normal condition. The phase is of 1-5 days in which the cavity of the Graafian follicle from which ovum had been expelled becomes reorganized and forms new structure known as corpus leteum (C.L.) which secretes progesterone hormone having following functions:

1. Prevents maturation of further Graffian follicles which prevent occurrence of further estrus period for a time.
2. It is essential for implementation of fertilized eggs.
3. It initiates the development of mammary gland.

Dioestrum: (Di-Estrus)

This is the longest phase of cycle. The corpus leteum is fully grown, making its effect on uterine wall to accommodate the embryo. The muscles of uterus develop. The uterine milk is produced to nourish embryo. If pregnancy succeeds, this stage is prolonged throughout gestation remaining C.L. intact for the period.

In absence of fertilized eggs, the C.L. undergoes retrogressive changes the cell becomes vaculated in the lipid droplets. Since the C.L. got reabsorbed, the level of progesterone is

.declined and the level of estradiol increases, bring the animal in heat and the cycle is repeated in case of failure of fertilization.

Pregnancy

The period from the date of conception to the day of parturition is called "gestation period" and the condition of the female of carrying the foetus during this period is called "Pregnancy".

OR

"The period of pregnancy is the duration of time which elapses between conception and parturition".

Importance of Pregnancy Diagnosis:

Whether animal is pregnant or not is directly related to economy of dairy management Pseudo-pregnancy may lead to loss of valuable time period in the life of animal.Pregnant animals need to change their feeding schedule as well as the management from early stage. An early detection of pregnancy becomes an indispensable job for herd owner.

Methods of Pregnancy Diagnosis:

1. Signs of Pregnancy - exhibited and and detected extrenally.
2. Symptoms of Pregnancy - per rectum / vaginum examination.
3. Laboratory Tests - Presence of certain hormones tested in laboratory.

A. Signs of Pregnancy:

1. Cessation of Oestrus cycle.
2. Sluggish temperament
3. Tendency to fatten.
4. Gradual drop in milk yield.
5. Gradual increase in weight
6. Increase in size of udder.
7. Waxy - appearance of teats in last month of pregnancy

B. Symptoms of Pregnancy:

This clinical diagnosis of pregnancy is most convenient and reliable method in which the examination of genitals can be done by the expert having adequate knowledge in anatomy & physiology of the livestock. The examinations can be done by two systems.

1. Per rectum examination of ovaries, uterus: The palpation of uterus per rectum, during early and mid-gestation periods, can draw positive conclusion by detecting characteristic changes that take place in uterus & uterine arteries.

Ovaries: The corpus leteum of pregnancy persists in ovary at its maximum size throughout gestation period. It is firm. Rounded at top & slightly elevated from surface of ovary. Ovary can be examined from 10 days of service up to 3 months of pregnancy.

2. per vaginum Examination: If vagina examination during pregnancy by means of a speculum, wall appears dry and wrinkled. During pregnancy the secretion of cervical glands becomes gelatinous & tough forming a plug for sealing the canal. The seal develops on 60th day.

C. Laboratory Test:

1. Use of ultrasonic devices: Tested with the help of devices working based on principles that sound waves develop due to heart beats. Foetal movement & reflected back at an altered frequency. It can be tested 30-35 days of pregnancy.

2. Progesterone assay: Progesterone level estimation by using radio immune assay can be done in both milk & plasma of pregnant animal with standards of non-pregnant animals.

3. Pattern of Vaginal Smear: In this method, vaginal smears stained & fixed and visible cells are classified and inference is made, however no popular method due to lack of accuracy.

4. Immunological Techniques: In this, serum from pregnant animals tested which shows containing a factor "early pregnancy factor" (EPF). EPF can be detected as early as six hours after fertilization. This is sensitive test with accuracy, early pregnancy can be detected.

5. Barium chloride test: When 5-6 drops of 1% barium chloride solution is poured to 5 ml. of urine clear white precipitate is formed in non-pregnant cows. While added to urine of pregnant cow, the increased content of estrogen and progesterone of urine prevent formation of any precipitate. Test is 95 - 100 % accurate, takes less than 3 minutes, give correct results 31-210 days after fertilization.

6. Pregnant mare serum test (PMS): This test is presently applicable to mares only. It is conducted by using 10 ml blood serum collected from mare between 50-85 days after fertilization. This serum injected into there vein of mature, non-pregnant female rabbit which has been isolated from all male rabbits for at least 30 days. Positive test shows dark red follicles in the ovaries of rabbit 48 hours after injection.

7. Scanning: The pregnancy can be easily diagnosed by the equipment called 'oviscan' in sheep, cattle, horses & dogs. The equipment helps to establish pregnancy within 30 days of fertilization in sheep.

Parturation

Definition:

Parturation is the expulsion of the foetus and its membranes from the uterus through the birth canal by natural forces and in such a state of development that the foetus is capable of independent life or in brief it is process of giving birth to a young one.

This process of cattle is called 'calving'. It is an absolutely normal physiological process.

Causes of Parturation:

The exact cause of Parturation is still a mystery but different prevalent views are there. These are summarized in brief as below:

A. Physical Factors:

1. **Progressive:** increase in irritability of uterus: Increase in size of foetus towards the period of gestation-enhances the irritability & sensitivity of uterus resulting of reflex expulsion of foetus.
2. **Distension of Uterus:** The action of extensive distension is followed by an equal & opposite reaction by uterus on foetus where it attempts to reduce to its original size thus expels the foetus.
3. **Infracts in Placenta:** All full term infracts are noticed in the placenta due to distension & consequent pressure on arteries. The blood supply is diminished & placenta becomes sensible. The nutrition of foetus interfered with it and becomes anarchic gaps which stimulates respiratory centre & concentration of uterine started.

4. **Fatty degeneration in Placentas:** During last stage of pregnancy, fatty degeneration of outer layer of placenta occurs resulting into separation of foetus. The foetus becomes foreign body & expelled out

B. Biochemical factors:

1. **Carbon dioxide tensions:** Accumulation of CO2 in blood occurs due to metabolic activities of foetus which sets uterine contractions.
2. **Exciting substances:** A full term foetus transmits certain substances to the maternal circulation due to insufficient nutrition. Believed to initiate Parturation.
3. **Antigen:** An excessive quantity of foetal antigen enters the maternal blood sream towards the end of pregnancy which interacts with existing liberated substances by blood antigens & initiates labour.

C. Hormonal Factors:

In total complex process, the known and unknown hormones from pituitary, ovary, adrenal, placenta, foetus & uterus act in coordinated manner but at end the estrogen level increases than progesterone making release of oxytocin which in turn initiates the contractions of uterus.

D. Neural Factors:

There is no evidence that functional relationship of the intrinsic innervations of uterus to its activity during labour. It is independents of center nervous system.

Stages of Parturation:

The act of Parturation is a continuous process but for the sake of understanding, the process is explained in four stages as:

1. **The preliminary stage:** Stage continuous for some hours to even days. External symptoms - swelling of udder, entire external genital organ becomes swollen & becomes reddish, clear, straw Coloured mucus secreted. The quarters droop/ slackening of muscles & ligaments of pelvic region. Behavior signs - animal looks for solitary place, cow feels uneasy, bellow and get excited.
2. **Dilation of Cervix Stage:** This lasts for 30 minutes to 3 hrs. The uneasiness increases and followed by labour pain, animal show signs of pain in its abdomen. It may lie and rise again several times. Labour pains become more acute with short intervals. The pulse quickened, breathing distressed and rapid. The water bag7 appear at vulva followed by fore feet of young one. At this time cervix is fully dilated.
3. **Expulsion of foetus stage:** It is period from complete dilation of the osuteri to the delivery of foetus. The back of cow arched, chest expanded, and muscles of abdomen became broad with each labour pain. At each contraction, water bag protrudes further from vulva till front hoof of young one is visible. Water bag bursts & fluid thrown off. When hoofs and nose are at genital, the head of calf is at pelvic which will have to pass through small pelvic opening; this moment is of supreme effort & greatest point of labour pain. At least, uterine contractions, combined with additional abdominal force on uterus, results in driving away the foetus through cervix, vagina & vulva.
4. **Expulsion of the after birth:** After expulsion of calf, the uterus tends to throw out the placental membrane which is now merely a foreign body. As a result of uterine contraction, the placenta separates from the cotyledons & passes into the vagina, where from it is expelled. Early expulsion of placenta is desirable to avoid putrefaction causing infection of uterus. Expulsion within 6-8 hrs is normal, beyond that manual removal is advised.

Care & Management of Cow (animal) before during and After Parturation (Calving)

Even though the Parturation is normal physiological process, it requires to take due care at all stages of Parturation by manager of the herd.

Before Parturation:

1. **Turning cow into a loose box:** To isolate from other animals, animal of advance pregnancy must be separated into calving box which must be cleaned & properly disinfected, bedded with clean, soft & absorbent litter
2. **Guarding Against Milk Fever:** In advanced pregnancy stage high yielding & first calvers are susceptible to Milk fever. To avoid it, provide enough minerals especially calcium by bone meal in daily diet. Give large doses of Vit. D about a week period to calving.
3. **Avoid Milking:** Prior to parturation which is likely to delay parturation by few hours.
4. **Watch for parturation signs:** Signs to know primary stage of parturation which are udder becomes large, dislended, herd, depressed or hollow appearance on either side of tail head, vulva enlarged in size, thick mucus discharge from valva, and uneasiness of the animal.

During Parturation:

1. **Dilation Phase:** Consists of the acts Le down & get ups, uneasiness due to labour pain, observe these acts from safe distance without making disturbances to animal.
2. **Parturation period:** In normal case period is of 2-3 hrs while in first calving 4-5 hrs or more Observe from safe distance without disturbing the animal.
3. **Watch for presentation of Calf:** The phase of expulsion of foetus, observe the appearance of water bag & its gradual emergence, bursting of it and appearance of fore feet with hoof & mouth.
4. **Normal presentation:** Any deviation from normal presentation of calf occurs; the immediate help of veterinarian should be taken being care of Dystokia.

After Parturation:

1. **Expulsion of placenta / after birth:** The placenta is discharged within 5-6 hrs. After calving in normal case while if not discharged within 6-7 hrs. Get the help of veterinarian and treat as per requirement.
2. Supply Luke-warm drinking water to cow.
3. When placenta expelled, prevent cow from eating.
4. The placenta should be properly disposed off by burying in ground.
5. Clean cow's body with clean & warm water with antiseptic.
6. Supply moistened bran with crude sugar or molasses.

Care with regard to milking of cow:

1. After Parturation when first milking, ensure that all blockages from teats removed.
2. Cow may be milked three times a day until the inflammation disappears from the udder.
3. Provide enough minerals i.e. calcium & phosphorus through diet & do not milk fully at a "time to avoid milk fever in high yielding cows

Care with regards to feeding:

1. Types of feeds provided - milk laxative, palatable &c nutritious.
2. Suitable feeds - Wheat bran, oats, and linseed oil seeds.
3. DCP & TDN of ration must be 16-18% & 70% respectively.
4. 40-60 gms. Sterilized bone meal & 40 gm common salt may be adder', to grains.
5. Succulent green, palatable fodders containing 50-60% legumes are suitable while amount concentrates should be increased gradually in three weeks.